

Non-Mendelian Pattern of Inheritance

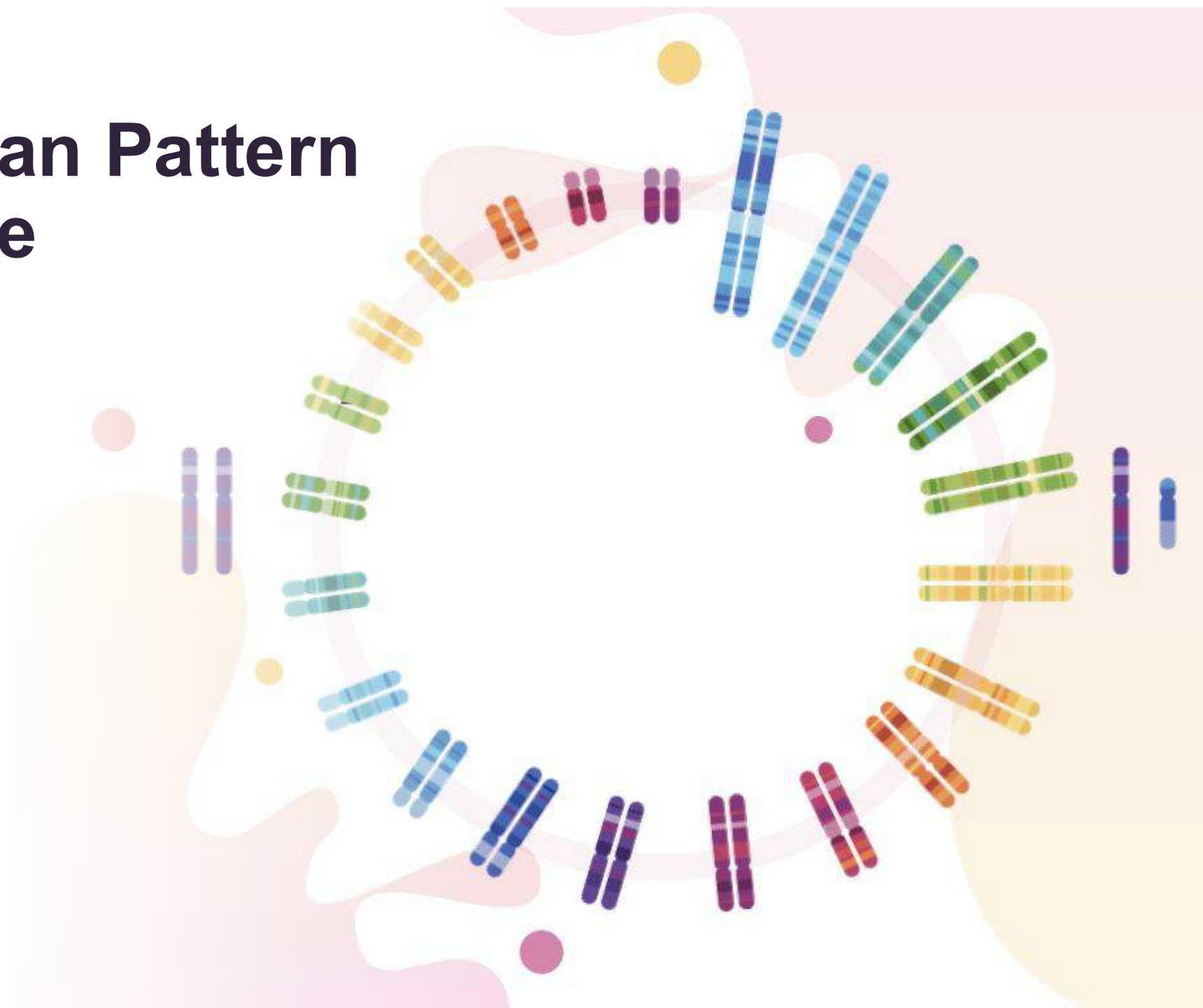
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Medicine, Faculty of
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2nd year medicine, 2026



Objectives

- Illustrate the various patterns of inheritance and bring them together with medical examples
- List the non-traditional patterns of inheritance
- State common clinical examples for non-traditional patterns of inheritance
- Recognize indications for referral to a genetic clinic.



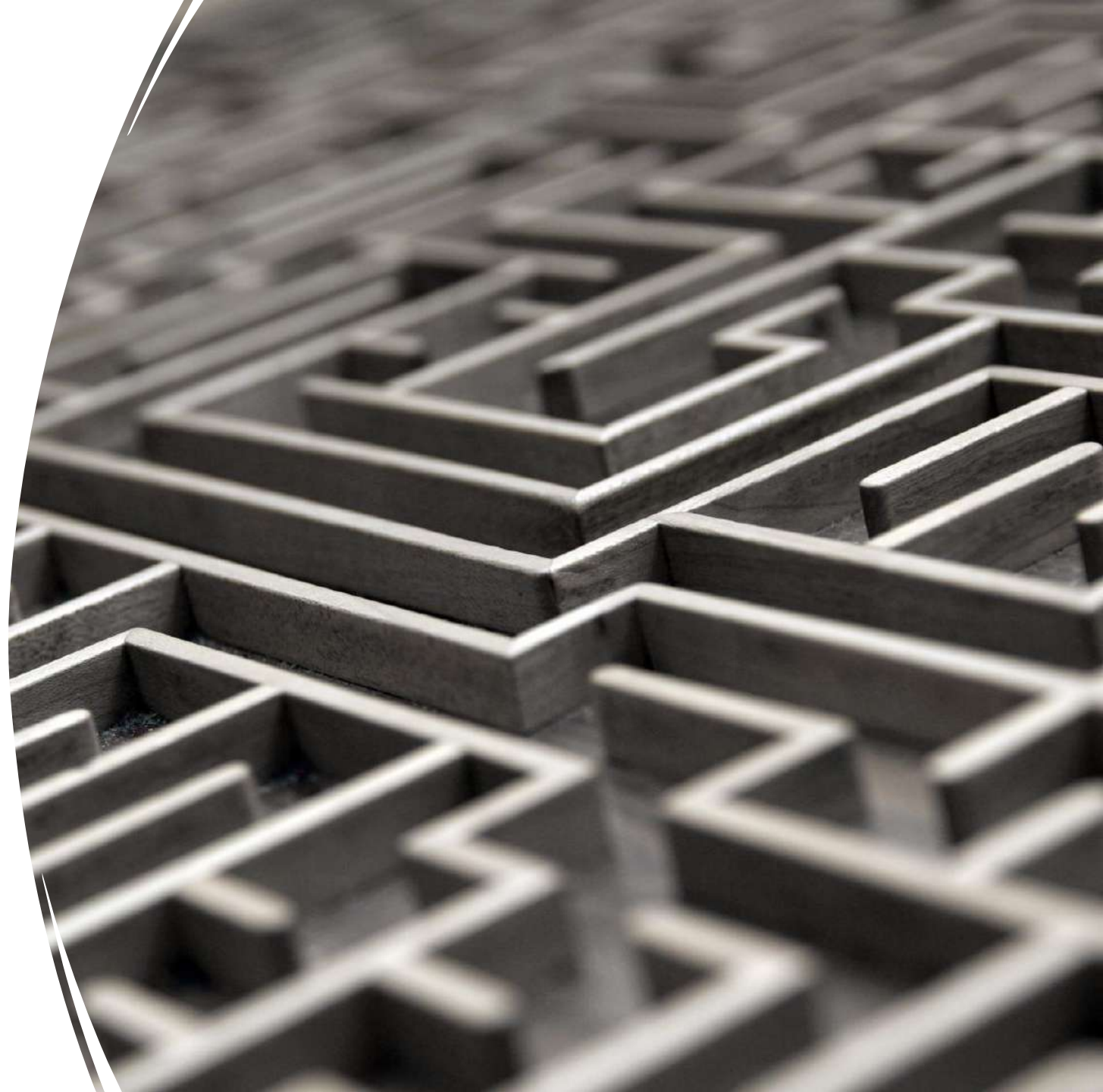
Outline: (Pages 290-310)

- Introduction
- Non-Mendelian (AKA non-classical) modes of inheritance:
 - Genomic Imprinting
 - Uniparental Disomy
 - Mitochondrial Inheritance
 - Anticipation
 - Mosaicism:
 - Somatic mosaicism
 - Gonadal mosaicism



Non-Mendelian Inheritance

- Non-Mendelian genetics → genetics that don't follow Mendel's laws
- Non-Classical Modes of Inheritance → several disorders are known which **do not follow** classical patterns of Mendelian inheritance
- Several different mechanisms have been recognized which can account for this



Non-Classical Modes of Inheritance

Genomic Imprinting

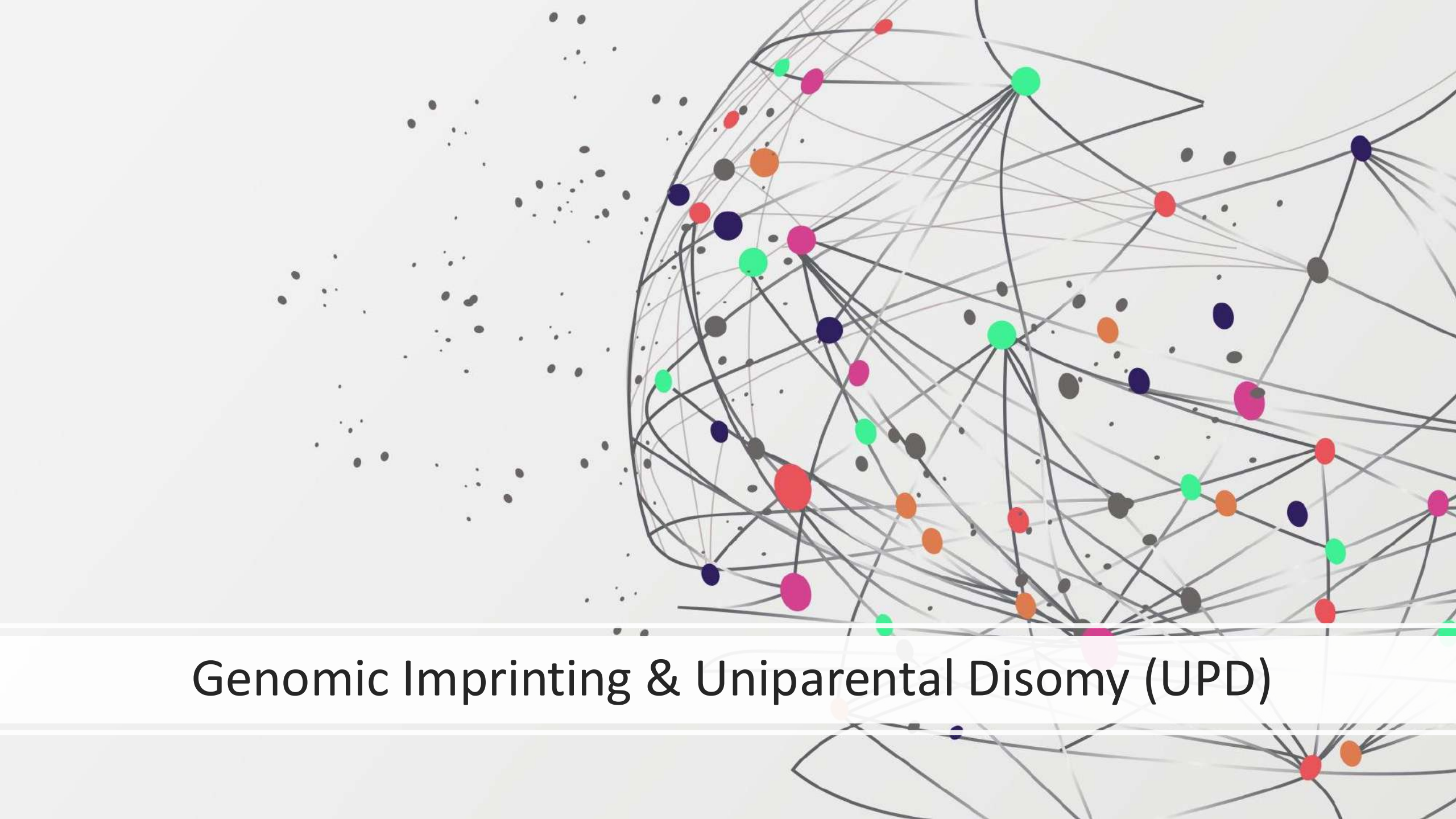
Uniparental Disomy

Mitochondrial Inheritance

Anticipation

Mosaicism:

- Somatic mosaicism
- Gonadal mosaicism

The background of the slide features a complex, abstract network diagram. It consists of numerous small, dark grey dots scattered across the light grey background. Overlaid on this are several larger, colored circular nodes in red, orange, yellow, green, blue, and purple. These nodes are interconnected by a dense web of thin, dark grey lines, creating a complex, interconnected pattern that resembles a molecular structure or a data network. The overall aesthetic is scientific and modern.

Genomic Imprinting & Uniparental Disomy (UPD)

Genomic Imprinting

- Genomic imprinting is an epigenetic phenomenon
- It was originally thought that genes on homologous chromosomes were expressed equally
- It is now recognized that different clinical features can result, depending on whether a gene is inherited from the father or from the mother
- This 'parent of origin' effect is referred to as genomic imprinting
- Methylation of DNA is the main mechanism by which expression is modified

Epigenetics and Methylation

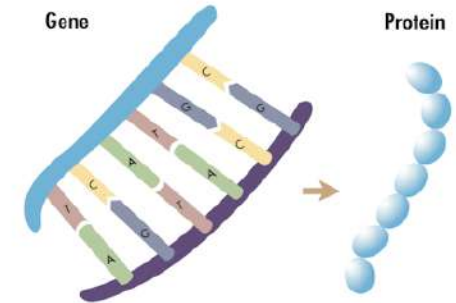
Epigenetics

The study of changes in gene function that are stable and heritable but do not involve a change in DNA sequence. One example is the process of DNA methylation.

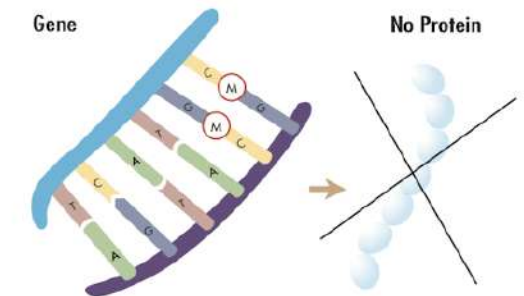
Methylation

The addition of a chemical (methyl group) usually to cytosine in DNA. This modification to DNA is important in regulating gene expression and in imprinting.

No Methylation



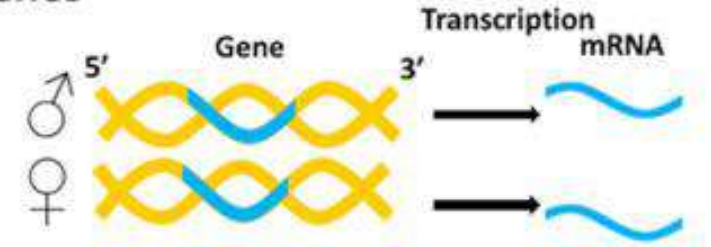
Methylation



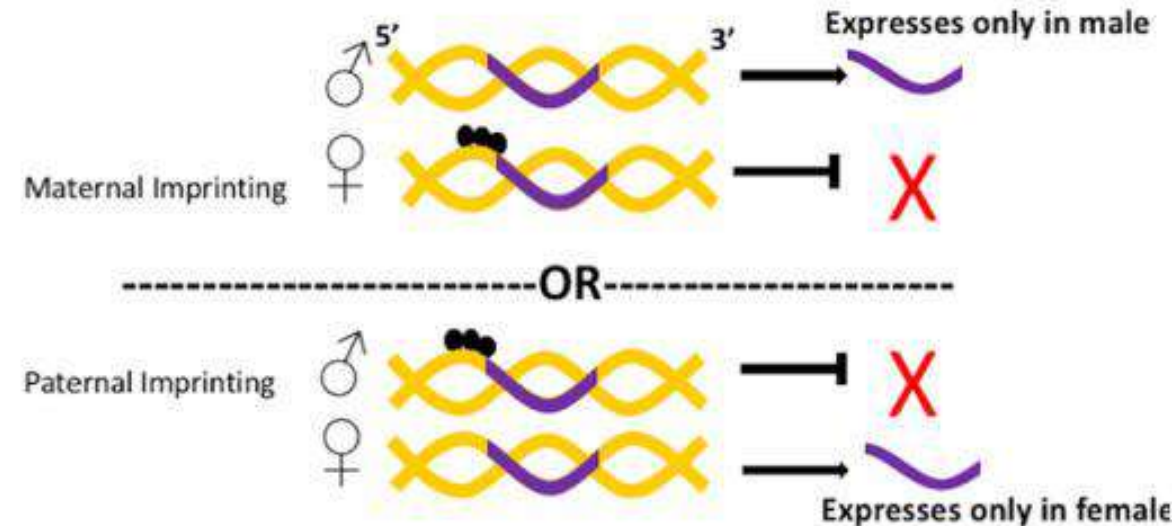
Genomic Imprinting

- At least 80 (& up to 100) human genes are known to be imprinted
- The differential allele expression (i.e., maternal or paternal) may occur in all somatic cells, or in specific tissues or stages of development
- The regions involved are known as differentially methylated regions (DMRs)
- These DMRs include imprinting control regions (ICRs) that control gene expression across imprinted domains

Non-imprinted genes



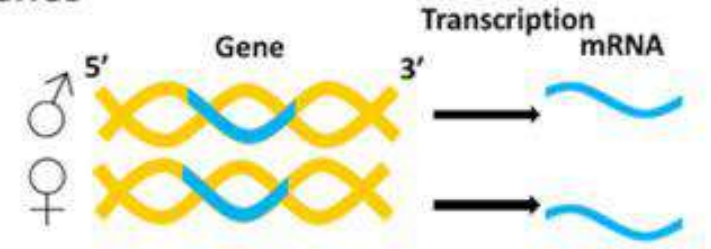
Imprinted genes



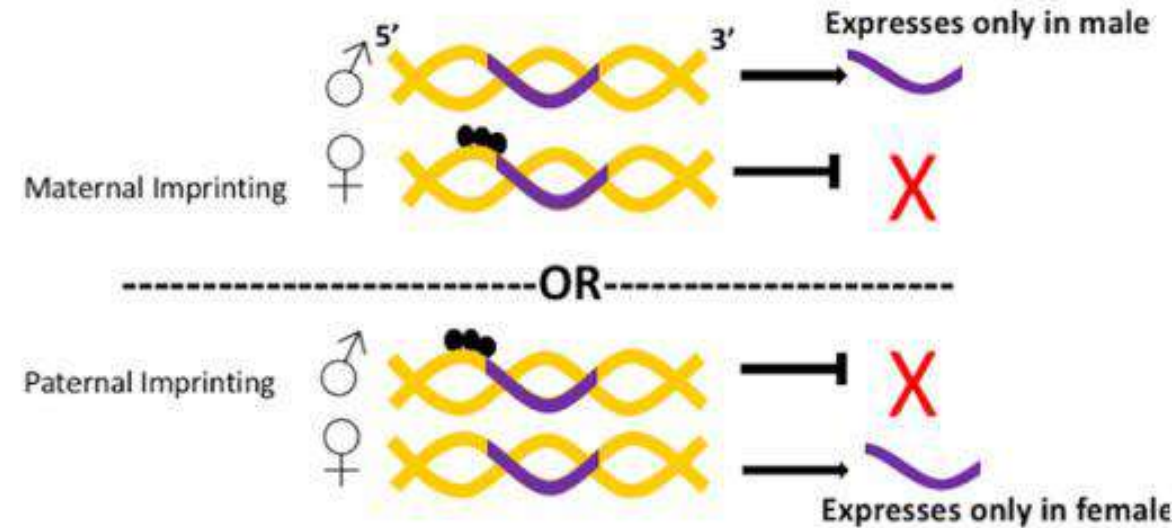
Genomic Imprinting

- For some human genes one of the alleles is transcriptionally inactive (no mRNA produced)
- This depends on from which parent the allele is received
- For example :
 - An allele transmitted by the mother is inactive
 - And the same allele transmitted by the father is active
 - The normal individual will have only one transcriptionally active copy of the gene
- This process of gene silencing is known as **IMPRINTING**
- The transcriptionally silenced genes are said to be **IMPRINTED**

Non-imprinted genes

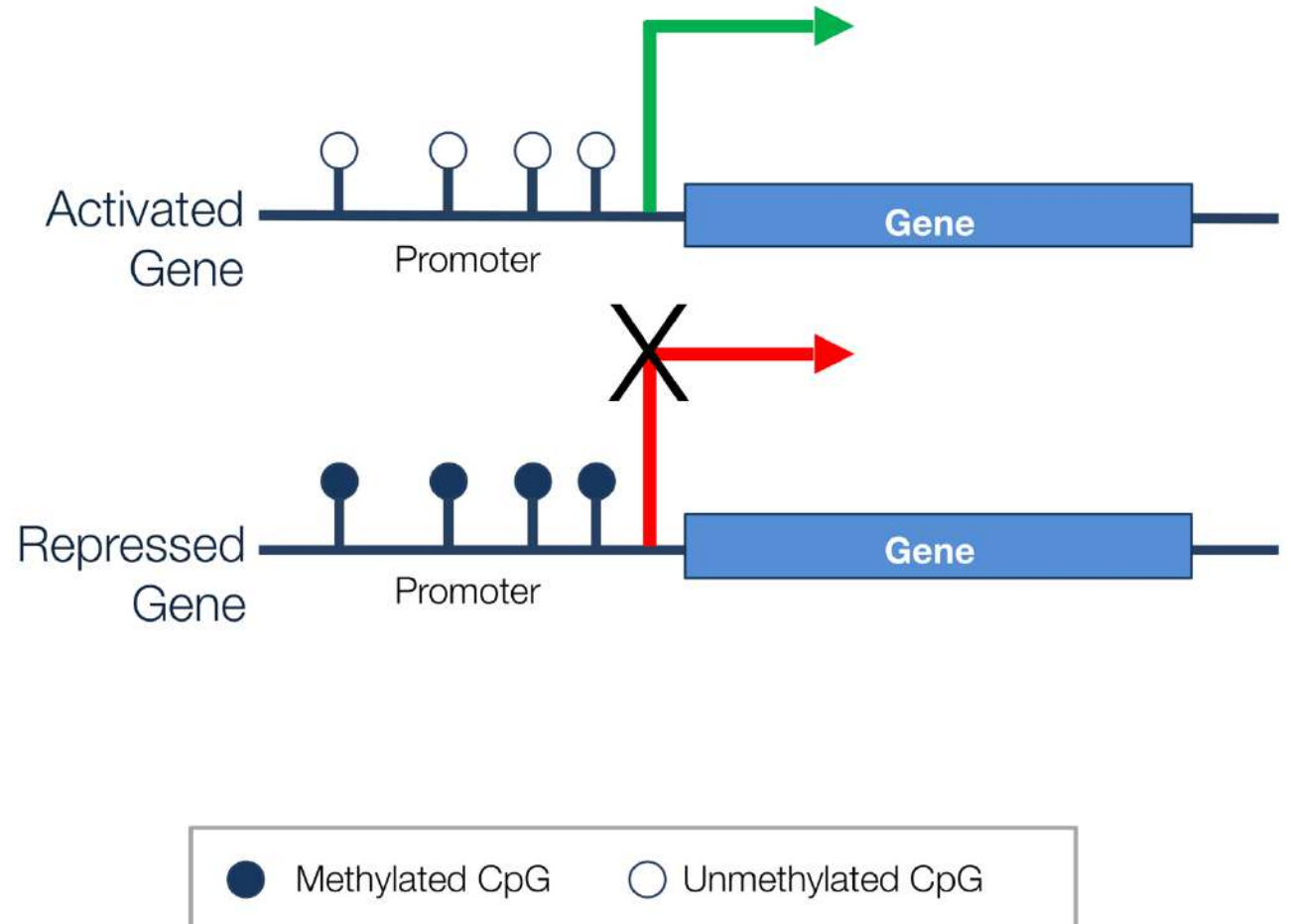


Imprinted genes



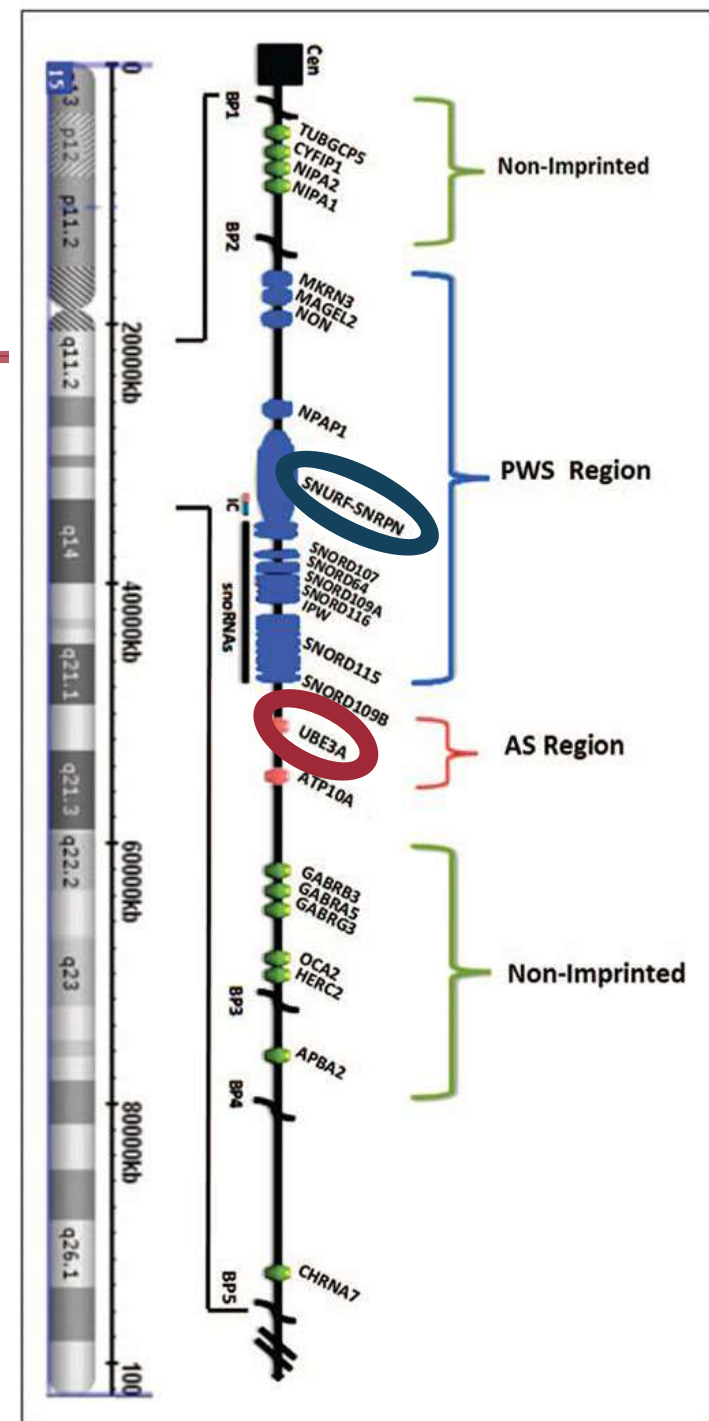
Genomic Imprinting

- Imprinted alleles tend to be **heavily methylated** in contrast to non-imprinted copy of the allele which typically not methylated
- The attachment of methyl groups to 5' region of genes inhibit the binding of proteins that promote transcription



Example: Chromosome 15 Disorders

- Evidence of genomic imprinting has been observed in two of well-known syndromes: Prader-Willi syndrome (PWS) and Angelman syndromes (AS) (chromosome 15q)
- **In PWS:** critical region contains several genes, including ***SNRPN gene***, that normally are transcribed (**active**) only on the chromosome 15 inherited from the **father**
- These genes are transcriptionally **inactive (IMPRINTED)** on the copy of chromosome 15 inherited from the **mother**
- On the other hand, **in AS:** other genes, including ***UBE3A gene***, in this critical region are transcribed (**active**) only on the chromosome inherited from the **mother** and are (**inactive**) on the chromosome inherited from the **father**



Uniparental Disomy

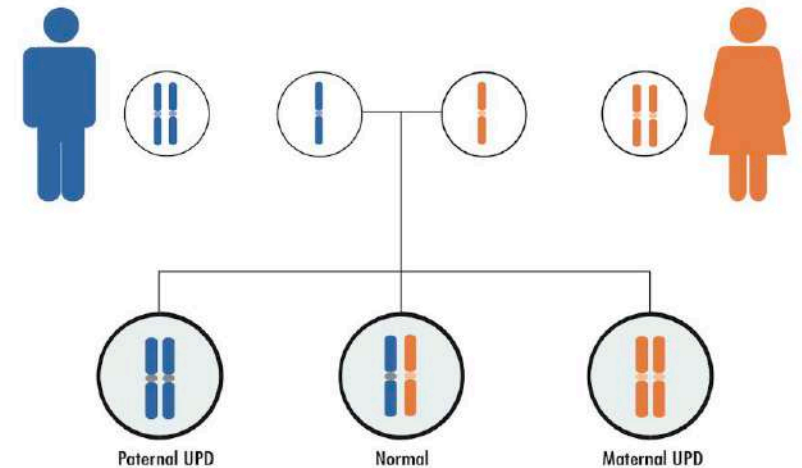


An individual normally inherits one of a pair of homologous chromosomes from each parent



If the individuals inherit both homologs of a chromosome pair from **only one parent**, it's called uniparental disomy (UPD)

Uniparental Disomy (UPD)



Mechanism of Origin of Uniparental Disomy

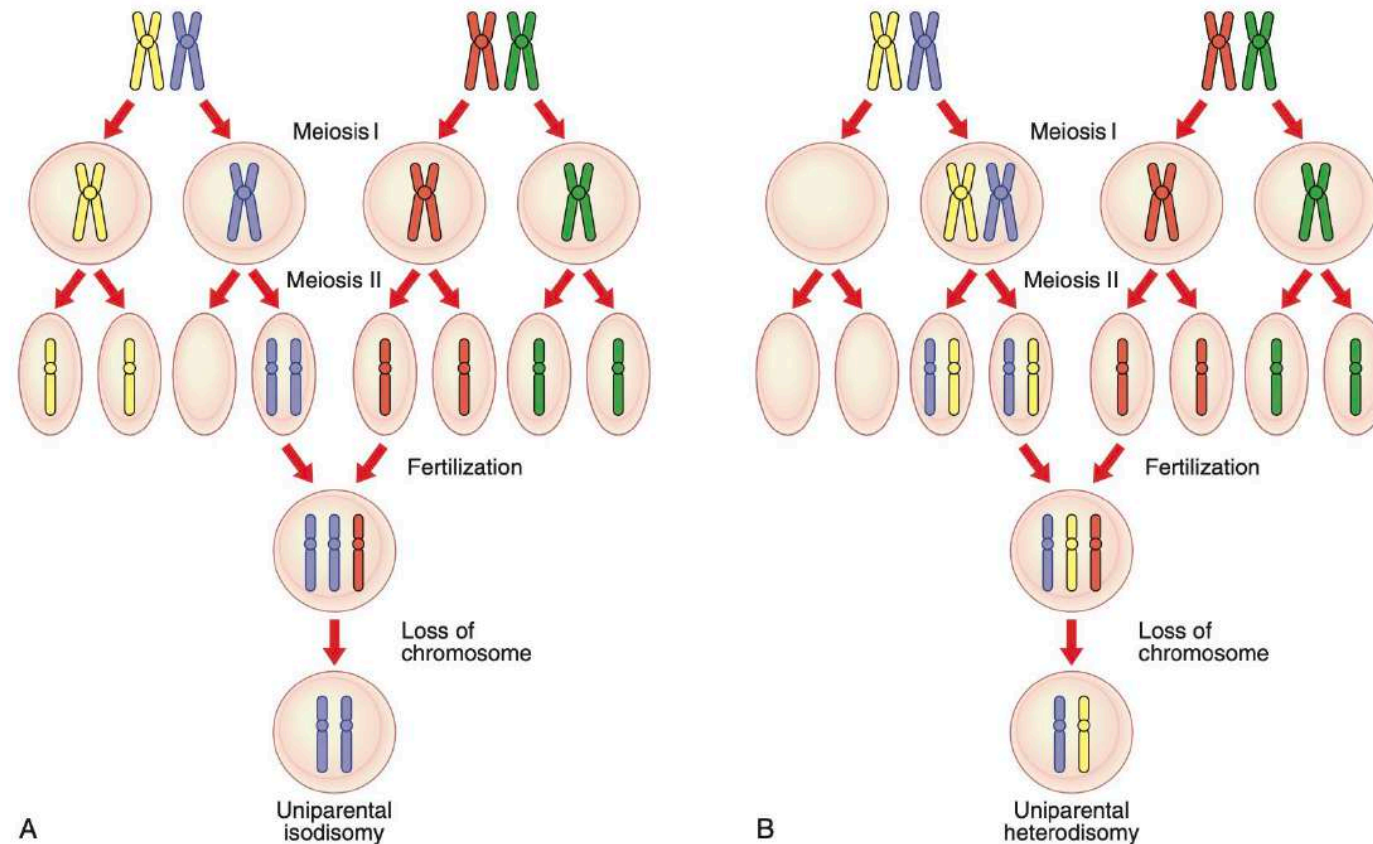
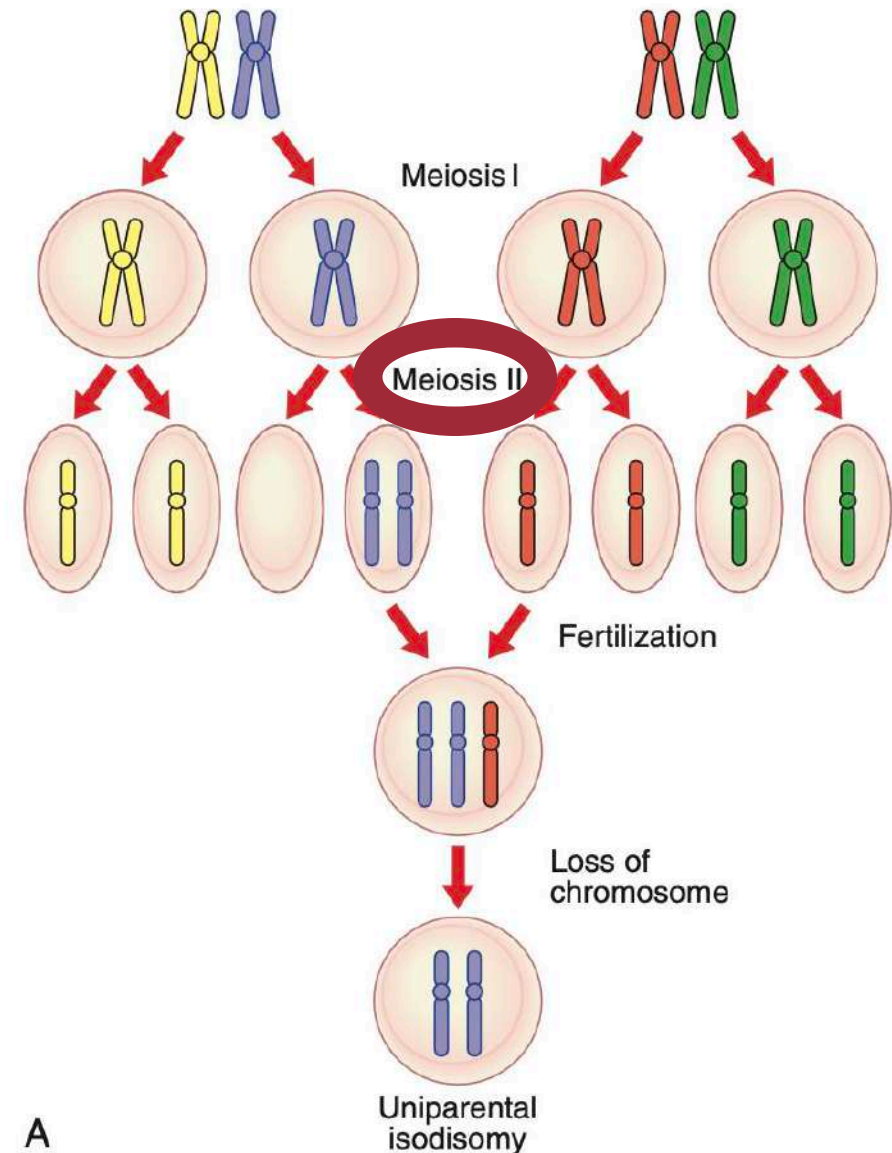


FIGURE 6.21 Mechanism of origin of uniparental disomy. **A**, Uniparental isodisomy occurring through a disomic gamete arising from non-disjunction in meiosis II fertilizing a monosomic gamete with loss of the chromosome from the parent contributing the single homolog. **B**, Uniparental heterodisomy occurring through a disomic gamete arising from non-disjunction in meiosis I fertilizing a monosomic gamete with loss of the chromosome from the parent contributing the single homolog.

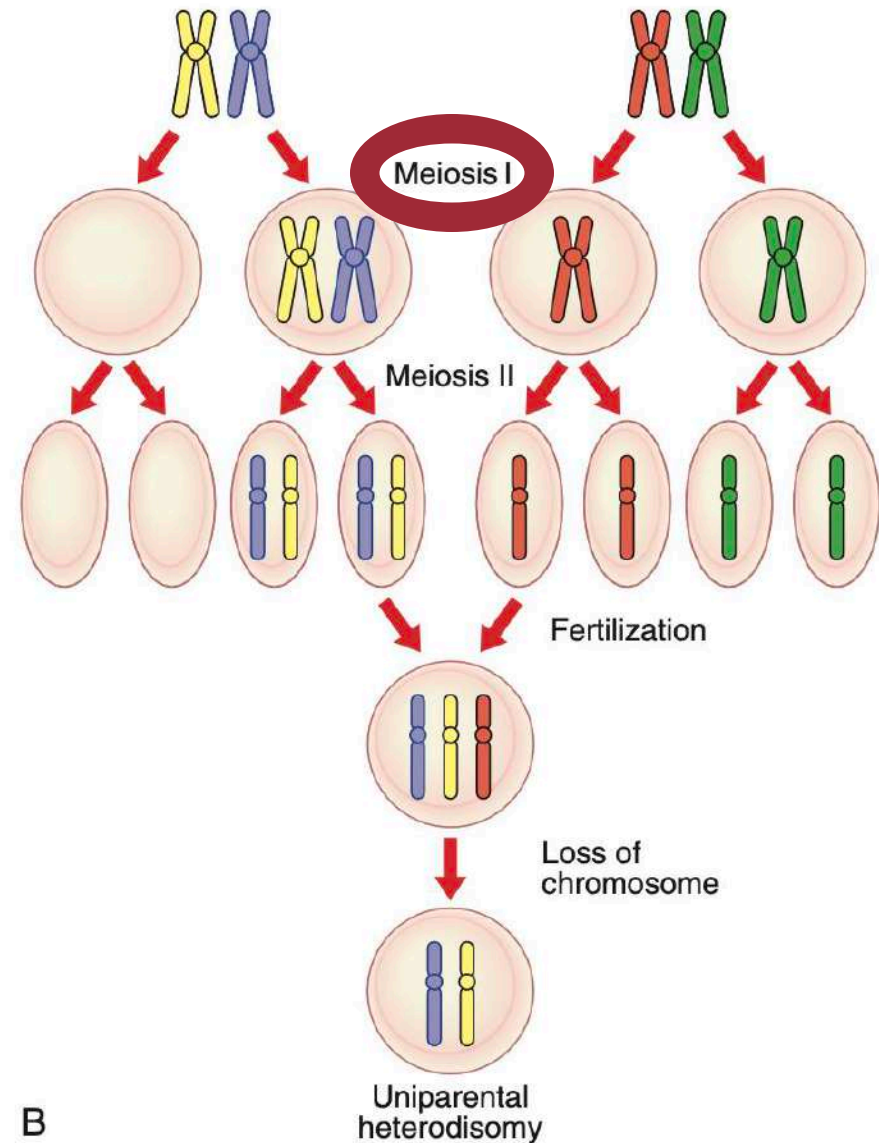
Mechanism of Origin of UPD: Uniparental Isodisomy

- If an individual inherits two copies of the same homolog from one parent, through an error in meiosis II → uniparental isodisomy
- A disomic gamete arising from non-disjunction in meiosis II → fertilizing a monosomic gamete → with loss of the chromosome from the parent contributing the single homolog



Mechanism of Origin of UPD: Uniparental Heterodisomy

- If an individual inherits the two **different** homologs from one parent through an error in **meiosis I** → uniparental heterodisomy
- A disomic gamete arising from **non-disjunction in meiosis I** → fertilizing a monosomic gamete → with loss of the chromosome from the parent contributing the single homolog



Uniparental Disomy

- If an individual inherits two copies of the same homolog from one parent, through an error in **meiosis II** → uniparental isodisomy
- If an individual inherits the two different homologs from one parent through an error in **meiosis I** → uniparental heterodisomy

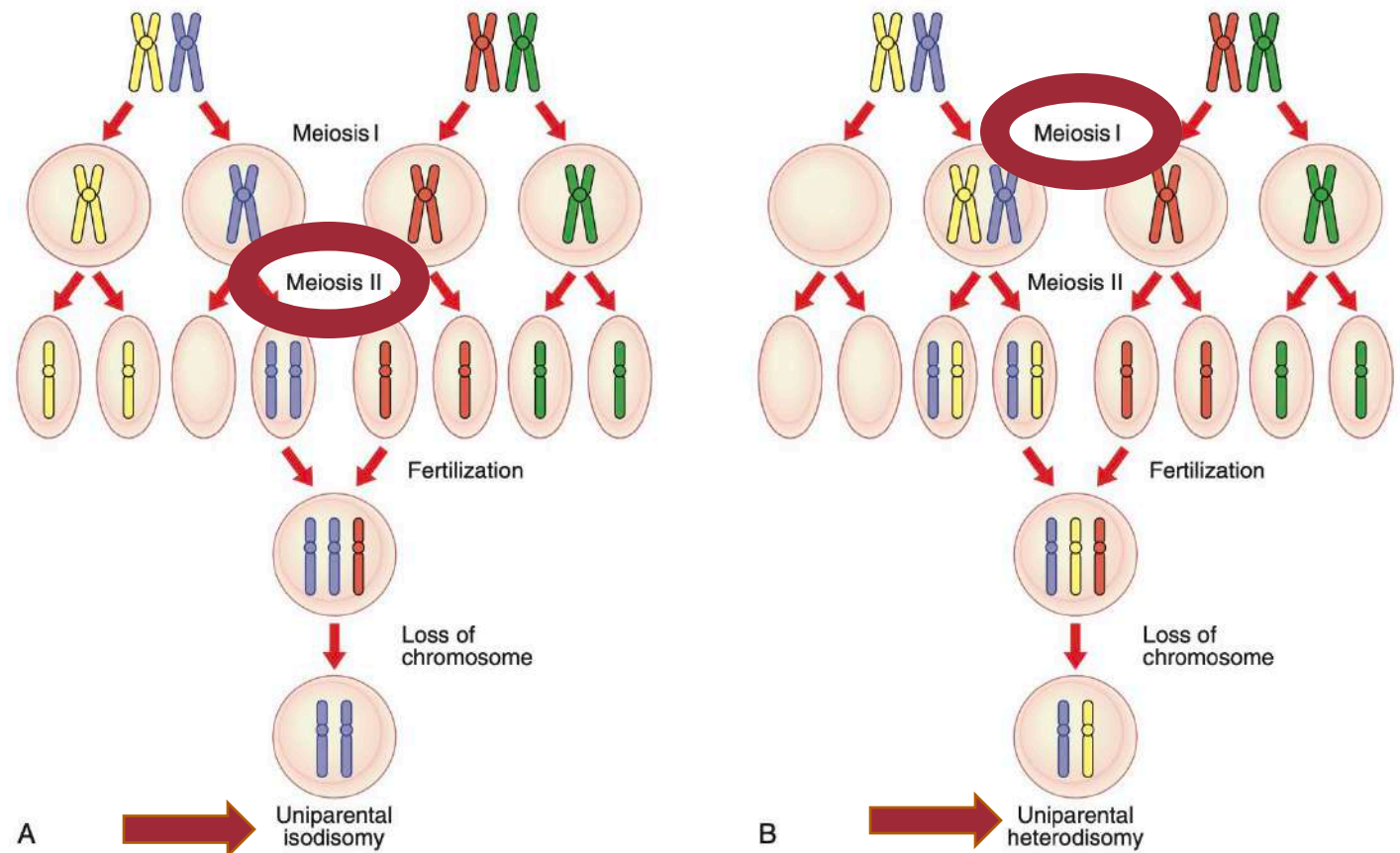


FIGURE 6.21 Mechanism of origin of uniparental disomy. **A**, Uniparental isodisomy occurring through a disomic gamete arising from non-disjunction in meiosis II fertilizing a monosomic gamete with loss of the chromosome from the parent contributing the single homolog. **B**, Uniparental heterodisomy occurring through a disomic gamete arising from non-disjunction in meiosis I fertilizing a monosomic gamete with loss of the chromosome from the parent contributing the single homolog.

Uniparental Disomy

- Several genes in this region are normally ACTIVE on only one chromosome
- If the single ACTIVE copy of one of these genes is lost, no gene product is produced at all, and disease results





Examples of Genomic Imprinting and UPD

Prader-Willi Syndrome

The genes associated with PWS normally are expressed only from a region of chromosome 15 inherited from the **FATHER** (Prader-Willi critical region, PWCR)

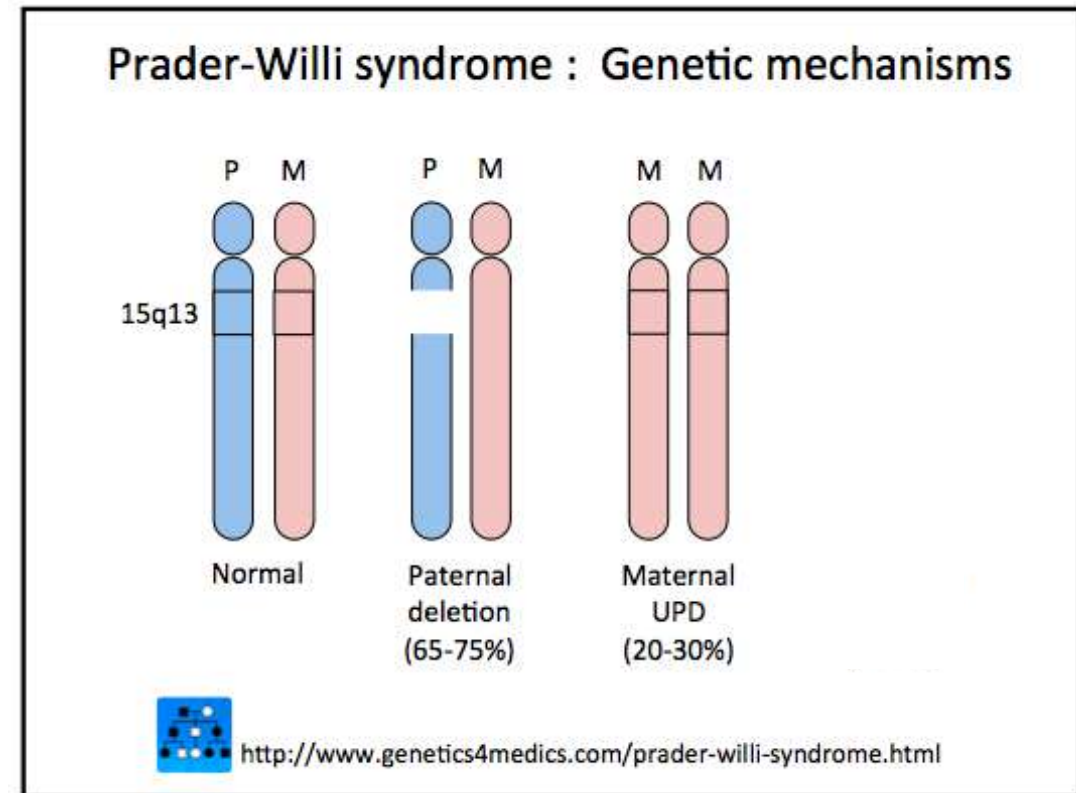
The genes inherited from the mother normally are inactivated

SNRPN gene encodes for small nuclear ribonucleoprotein polypeptide N protein, involved in pre-mRNA splicing

Prader-Willi Syndrome: Genetic Mechanism

Caused by loss of the paternally (FATHER) contributed PWS critical region of chromosome 15(q11-q13) caused by:

1. Deletion of the **paternally** inherited 15q11-q13 region (chromosome from the FATHER)
2. Uniparental disomy of the **maternal** chromosome 15q11.2-q13 region (UPD 15) (inherited two copies of this chromosomal region from the mother (chromosome of the FATHER is lost))



Prader-Willi syndrome

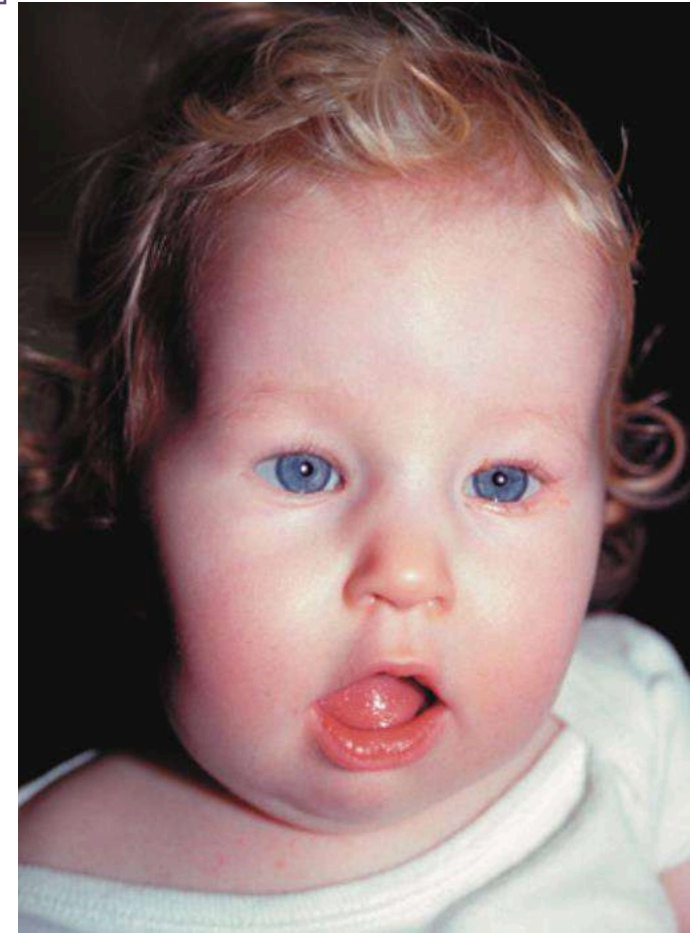
Cytogenetic locus: 15q11-13

Clinical features:

- Neonatal hypotonia
- Failure to thrive in infancy, then hyperphagia leading to **obesity**
- **Short stature**, small hands and feet
- **Hypogonadism** with cryptorchidism
- Dysmorphic features: almond-shaped palpebral fissures, narrow nasal bridge, thin upper lip with down-turned corners of the mouth
- Developmental delay (DD), intellectual disability (ID), learning difficulties, and behavioral problems

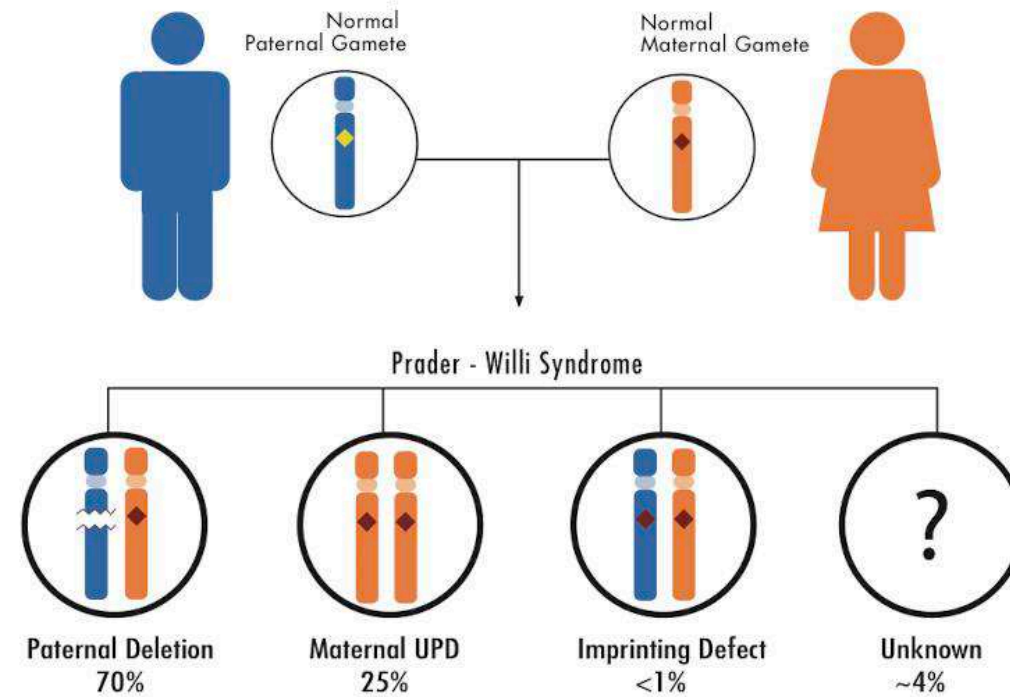
Inheritance:

- Up to 70% → paternal deletion of 15q11.2-q13
- 25% to 30% → maternal uniparental disomy of chr 15



Inheritance of Prader-Willi Syndrome

- ◆ Active PWS-related genes
- ◆ Inactive PWS-related genes



Adapted from Journal of the American Academy of Child and Adolescent Psychiatry, 2000;39:388



Angelman syndrome

The genes associated with AS normally are expressed only from a region of chromosome 15 inherited from the **MOTHER**

The genes inherited from the FATHER normally are inactivated

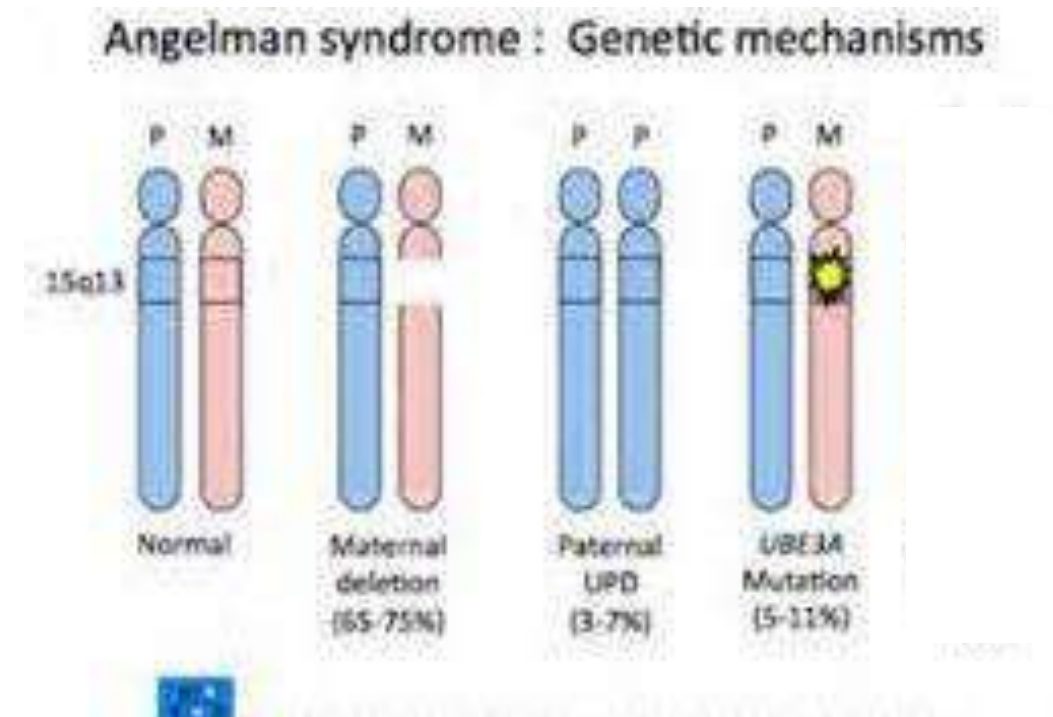
UBE3A gene provides instructions for making a protein called ubiquitin protein ligase E3A protein

This enzyme is involved in targeting proteins for degradation within cells

Angelman Syndrome: Genetic Mechanism

Caused by loss of the maternally (MOTHER) contributed AS critical region of chromosome 15(q11-q13) caused by:

1. Deletion of the **maternal** inherited 15q11-q13 region, which includes *UBE3A* gene (chromosome from the MOTHER)
2. Uniparental disomy of the **paternal** chromosome 15q11.2-q13 region (UPD 15) (inherited two copies of this chromosomal region from the father (chromosome of the MOTHER is lost))
3. Mutation/pathogenic variant in the **maternally** derived *UBE3A* gene



Angelman Syndrome

Responsible gene: *UBE3A*

Cytogenetic locus: 15q11-q13

Clinical features:

- Severe developmental delay or ID, severe speech impairment
- Microcephaly and seizures, and unsteady/ataxic gait
- Inappropriate happy demeanor that includes frequent laughing, smiling, and excitability

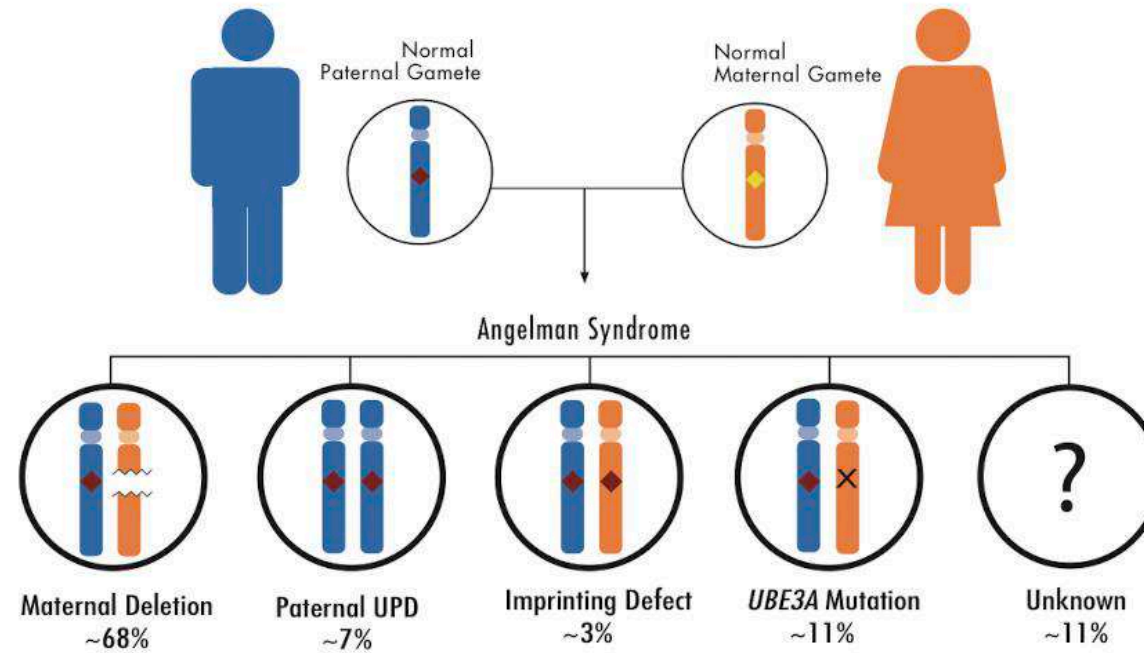
Inheritance:

- 70% → maternal deletion of 15q11.2-q13
- 5% → paternal uniparental disomy of chr 15
- 10% → *UBE3A* pathogenic variants



Inheritance of Angelman Syndrome

- ◆ Active AS-related gene (*UBE3A*)
- ◆ Inactive AS-related gene (*UBE3A*)
- × *UBE3A* Mutation

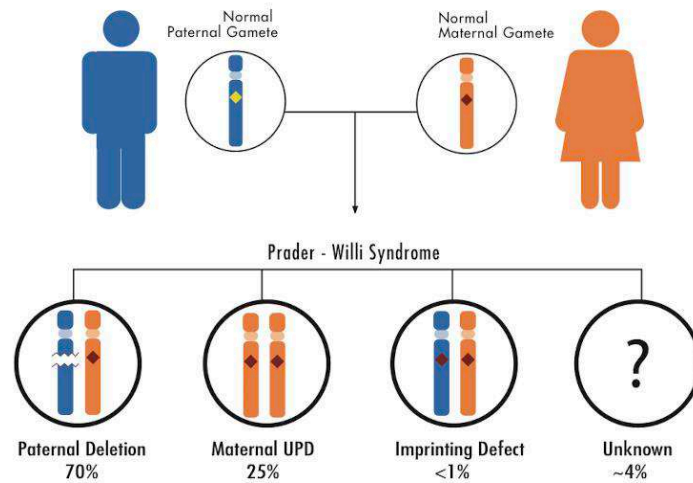


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Inheritance of Prader-Willi Syndrome

- ◆ Active PWS-related genes
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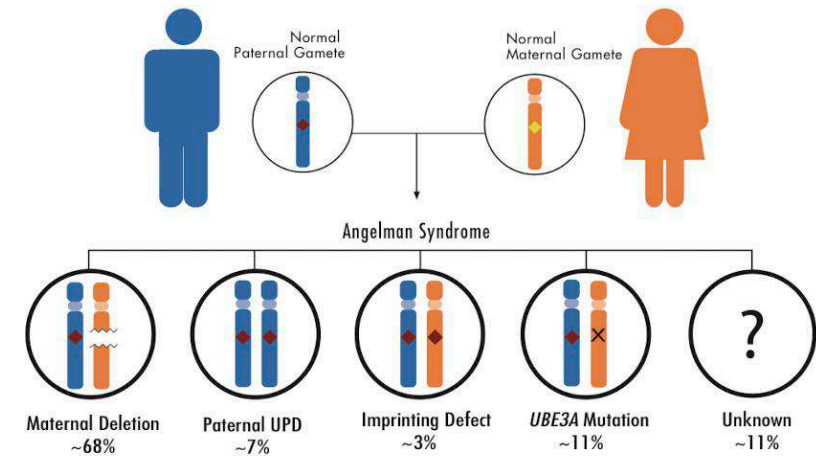


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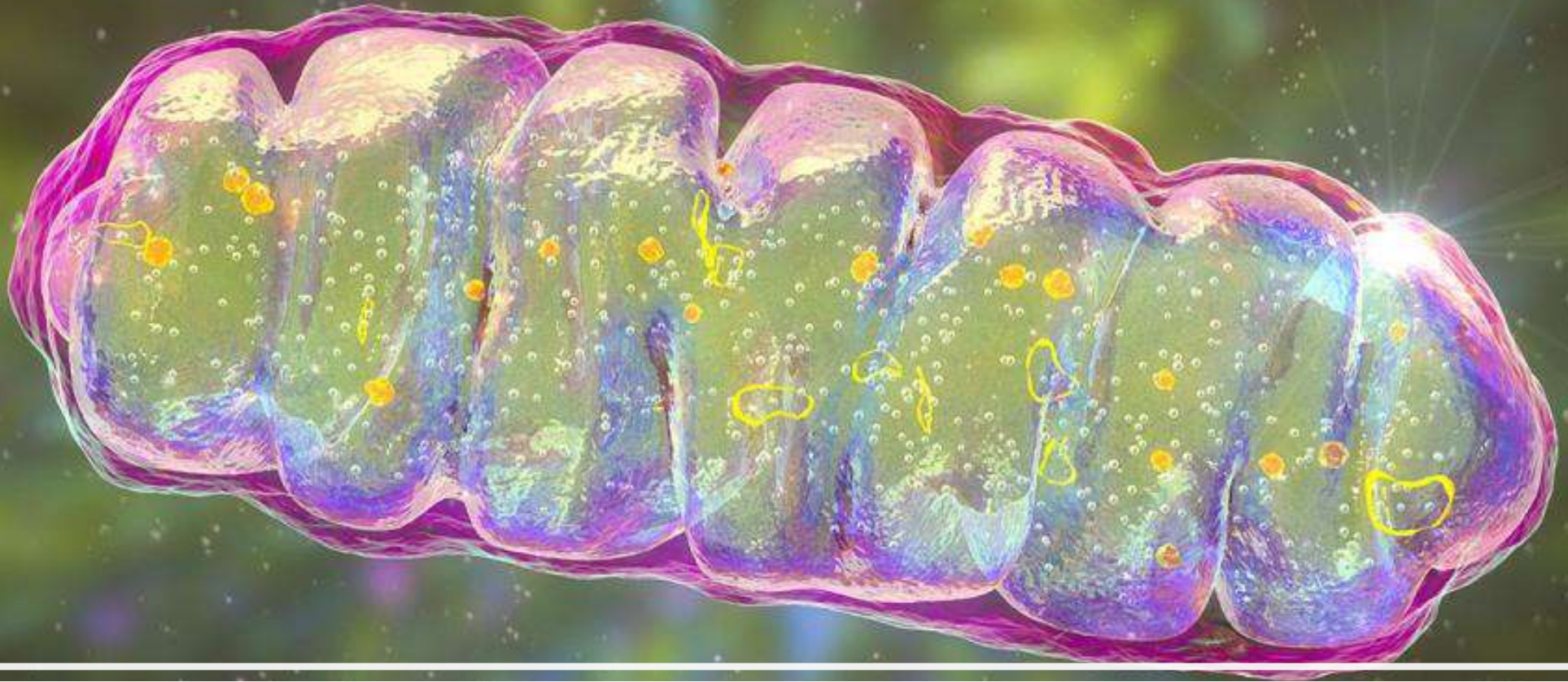
Inheritance of Angelman Syndrome

- ◆ Active AS-related gene (*UBE3A*)
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- × *UBE3A* Mutation



Adapted from Journal of the American Academy of Child and Adolescent Psychiatry, 2000;39:388

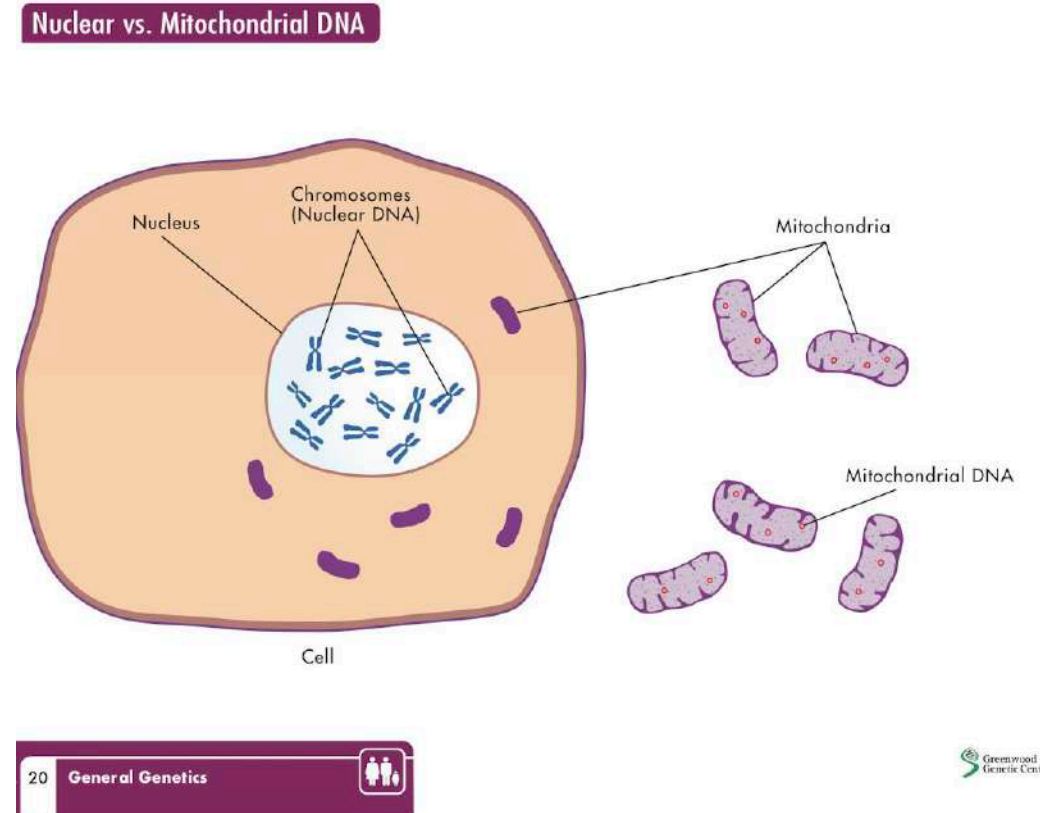




Mitochondrial Inheritance

Mitochondrial Inheritance

- The great majority of the genetic diseases caused by defects in the nuclear genome (nuclear DNA)
- However, a small but significant number of the diseases can be caused by mutation in mitochondrial DNA (mtDNA)
- Each cell contains thousands of copies of mitochondrial DNA with more being found in cells that have **high energy requirements**, such as brain and muscle
- Mitochondrial DNA has a **HIGHER** rate of spontaneous mutation than nuclear DNA
- The accumulation of mutations in mtDNA has been proposed as being responsible for some of the somatic effects seen with aging



Mitochondrial Inheritance

- Mitochondria, and their DNA, are inherited almost exclusively from the mother through the oocyte
- Mitochondrial inheritance explains the pattern of inheritance observed in some rare disorders that affect both males and females but are transmitted only through females, so-called **MATERNAL OR MATRILINEAL INHERITANCE**

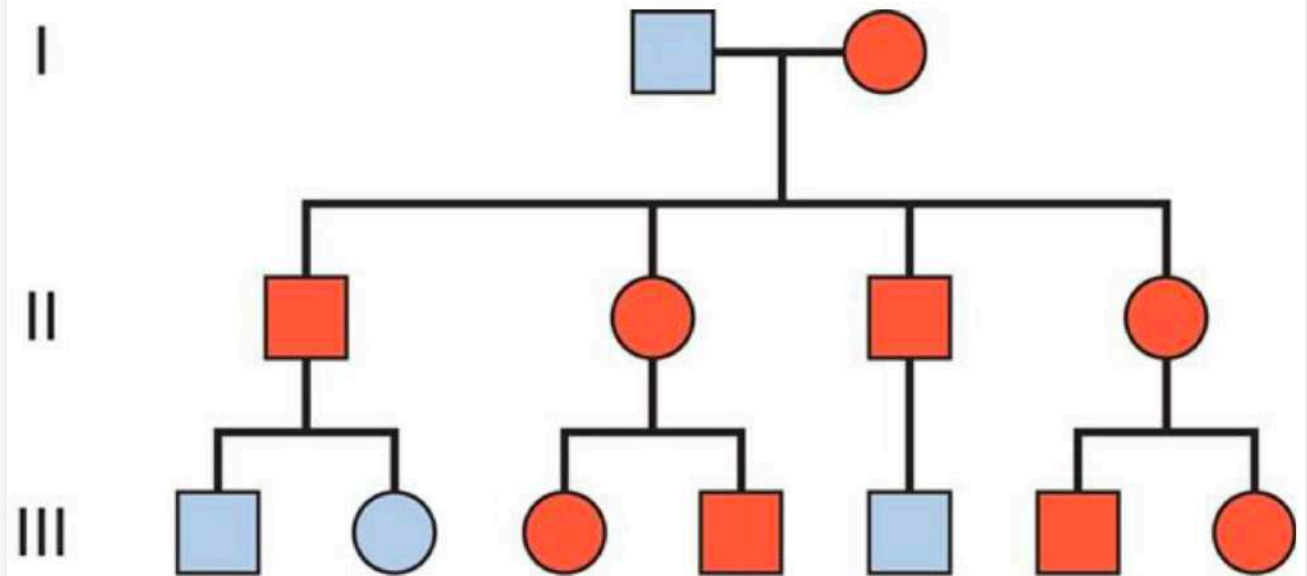


FIG. 6.29 Family tree consistent with mitochondrial inheritance.

Mitochondrial Inheritance

- In most persons, the mitochondrial DNA from different mitochondria is identical → termed **homoplasmy**
- If a variant occurs in the mitochondrial DNA of an individual, there will be two populations of mitochondrial DNA → termed **heteroplasmy**
- The proportion of mitochondria with a variant in their DNA varies between cells and tissues, and this, together with variant heterogeneity, explains the range of phenotypic severity seen in persons affected with mitochondrial disorders
- The disease may be **mild, moderate or severe** according to the distribution and amount of abnormal mitochondria in the affected cells and tissues

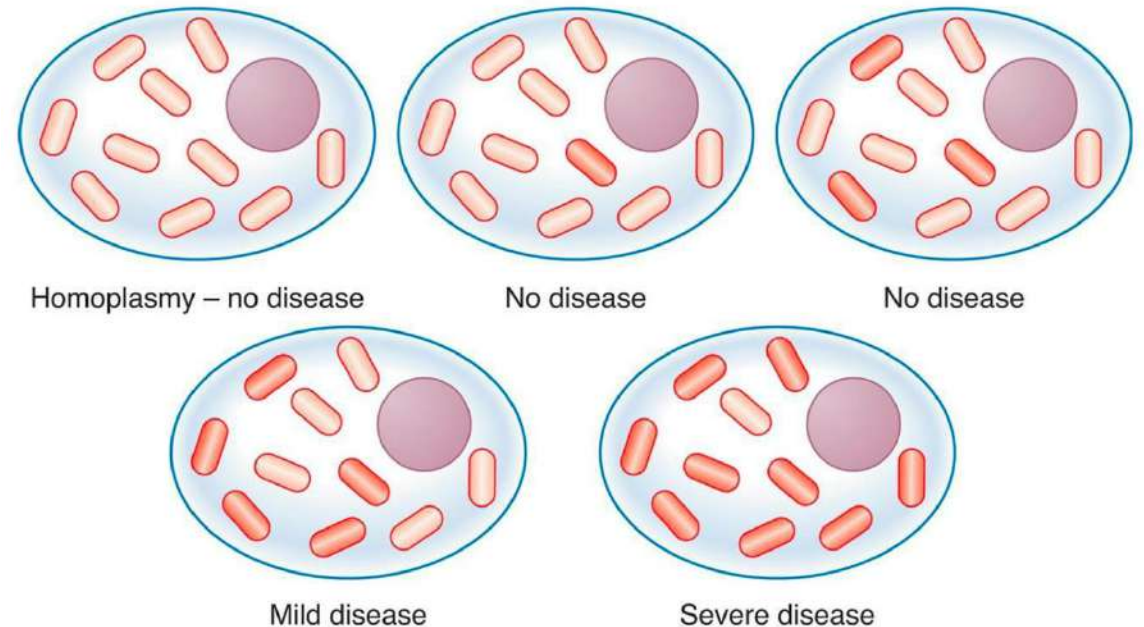
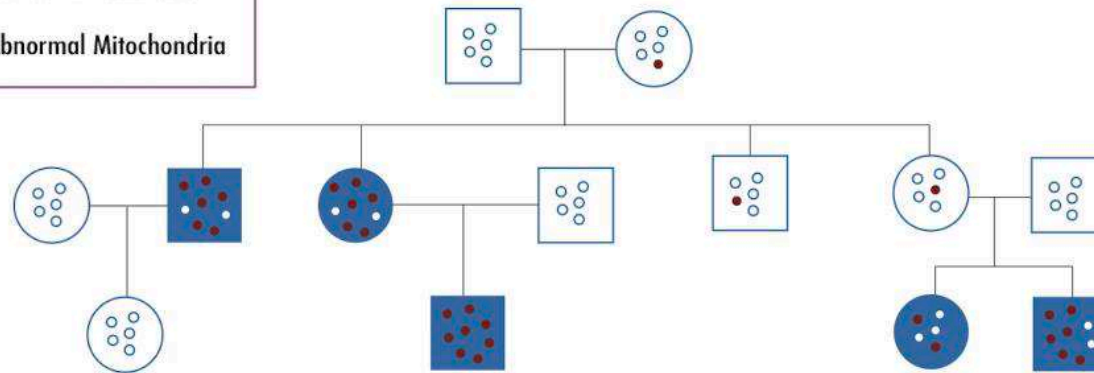
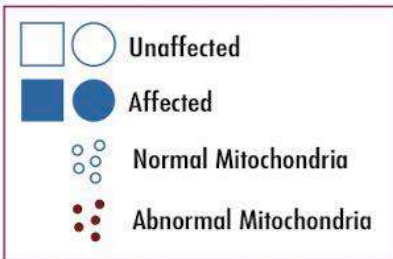


FIG. 6.30 Progressive effects of heteroplasmy on the clinical severity of disease from variants in the mitochondrial genome. Low proportions of variant mitochondria are tolerated well, but as the proportion increases, different thresholds for cellular, and hence tissue, dysfunction are breached (mauve circle represents the cell nucleus).

Mitochondrial Inheritance



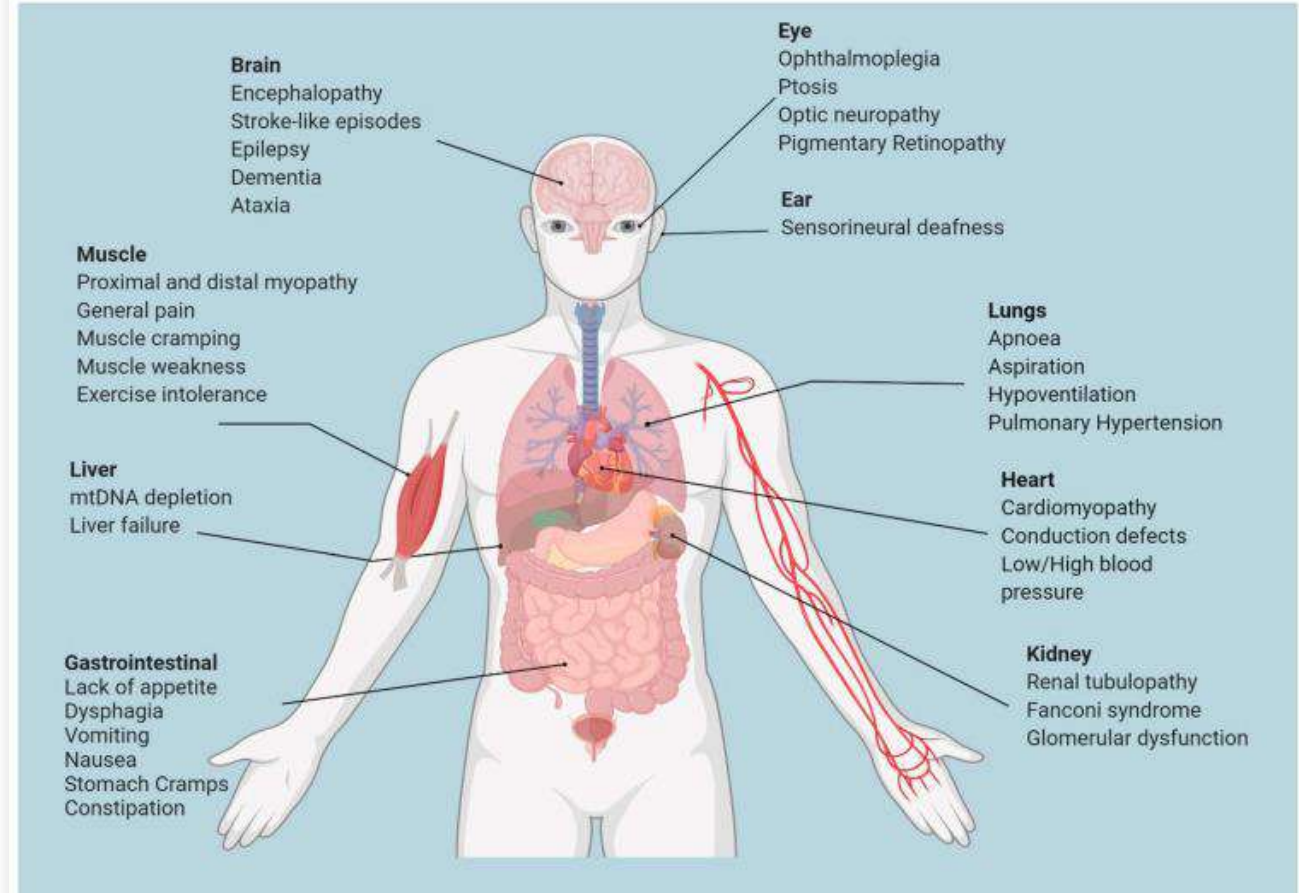
Characteristics of Mitochondrial Inheritance

- Mitochondria are inherited from the mother.
- All offspring of an affected or carrier female are at risk of being affected.
- All daughters of an affected or carrier female are at risk of transmitting the condition.
- Affected males cannot pass the condition to any of their children.



Clinical Manifestation of Mitochondrial Disorders

- Unusual combinations of neurological and myopathic features.
- May occur with:
 - Cardiomyopathy and conduction defects, diabetes and deafness
- **Pathophysiology:** Mitochondria drive cellular metabolism via oxidative phosphorylation.
- Most affected organs:
 - **Central nervous system (CNS)**
 - **Skeletal muscle**
 - **Heart**
- **Examples:**
 - **MELAS:** Mitochondrial encephalopathy, lactic acidosis, stroke-like episodes
 - **MERRF:** Myoclonic epilepsy with ragged red fibers



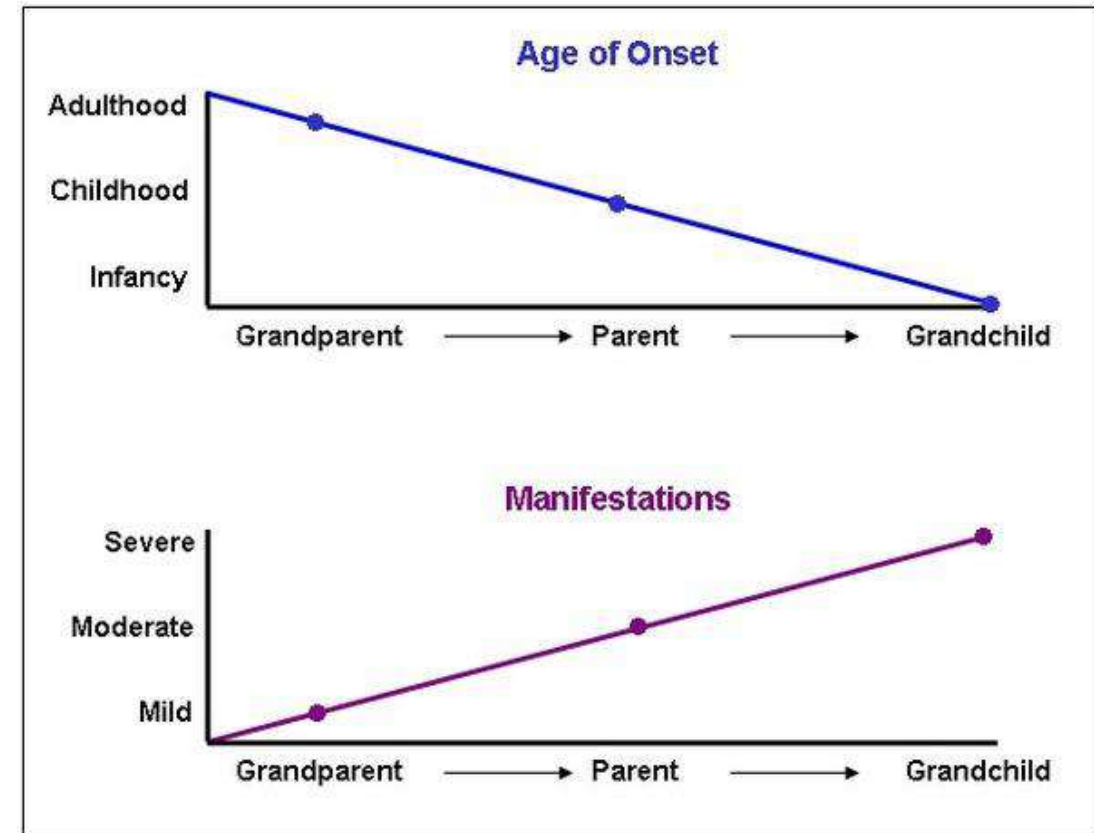
Di Donfrancesco A, Massaro G, Di Meo I, Tiranti V, Bottani E, Brunetti D. Gene Therapy for Mitochondrial Diseases: Current Status and Future Perspective. *Pharmaceutics*. 2022 Jun 17;14(6):1287. doi: 10.3390/pharmaceutics14061287. PMID: 35745859; PMCID: PMC9231068.

The background is a dark green field filled with numerous overlapping circles of various sizes and colors, including shades of purple, blue, green, yellow, and red. A horizontal white bar spans the width of the image, containing the text "Anticipation & Repeat Expansion".

Anticipation & Repeat Expansion

Anticipation

- If the onset of the disease occurs at earlier age in the offspring than in the parents
 - And/or the disease occurs with increasing severity in subsequent generations
- this phenomenon is called anticipation
- Anticipation occurs as a result of the expansion of unstable DNA triplet repeat sequences





Repeat Expansion

Trinucleotide Repeat Disorders

Trinucleotide **C G G**

• A group of 3 chemical bases within a gene.

Normal number of repeated trinucleotide segments within a gene = Normal Protein Function.



Mildly expanded number of repeated trinucleotide segments = Normal Protein Function.



Fully expanded number of repeated trinucleotide segments = Abnormal or Absent Protein.



- Nucleotide repeats are short DNA sequences that are repeated a number of times in a row
- For example:
 - Trinucleotide repeat is made up of 3-base-pair sequences (CGG)
- A repeat expansion is a dynamic mutation/variant that increases the number of times that the short DNA sequence is repeated
- This type of variant can cause the resulting protein to function improperly



Anticipation & Repeat expansion

+ Expansion of DNA repeats has been shown to cause anticipation in some genetic diseases

○ Examples: (Table 2.5, Page 102)

- Huntington disease (due to **CAG repeats**)

Normal range 9-35 repeats

Disease range : full penetrance more than 40 repeats with early onset, (36-39) repeats with reduced penetrance and late onset

- Myotonic dystrophy (due to **CTG repeats**)

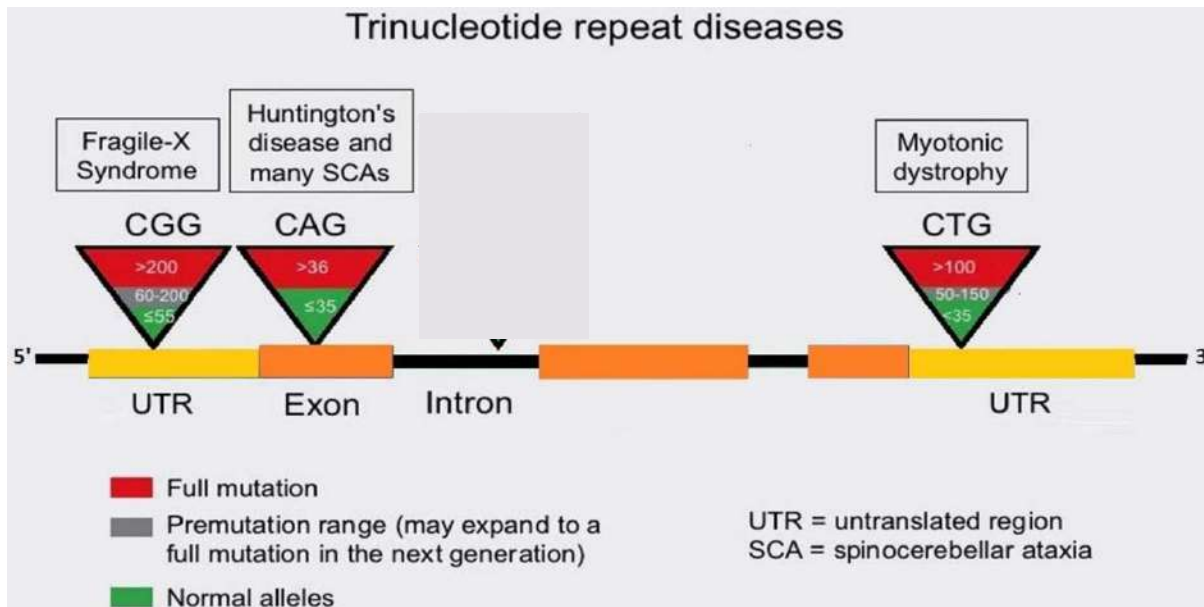
Normal range 5 – 35 repeats

Disease range 50 up to several 1000 repeats

- Fragile X-syndrome (due to **CGG repeats**)

Normal range 10 – 50 repeats

Disease range 200 – 2000 repeats



Bhowmik, A. D., Rangaswamaiah, S., Srinivas, G., & Dalal, A. B. (2015). Molecular genetic analysis of trinucleotide repeat disorders (TRDs) in Indian population and application of repeat primed PCR. *European Journal of Medical Genetics*, 58(3), 160-167.

Table 2.5, Page 102

Table 2.5 Examples of diseases arising from repeat expansions

Disease (Gene)	Repeat Sequence	Normal Range (Repeats)	Pathogenic Range (Repeats)	Repeat Location
Huntington disease (<i>HTT</i>)	CAG	9–35	36–100	Coding
Myotonic dystrophy type 1 (<i>DMPK</i>)	CTG	5–35	50–4000	3' UTR
Fragile X site A (<i>FMR1</i>)	CGG	10–50	200–2000	5' UTR

Fragile X Syndrome (FXS) *Page 912*

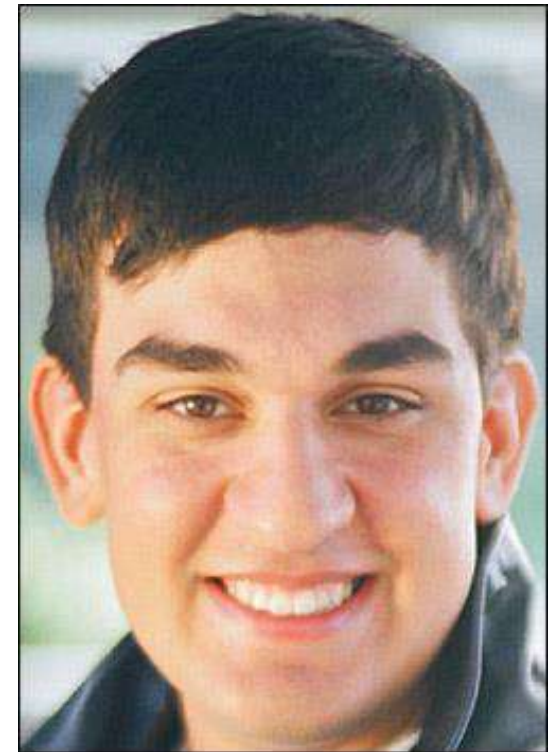
Responsible gene: *FMR-1* (Fragile X Mental Retardation Protein), plays a crucial role in the development and function of cerebral neurons

Cytogenetic locus: Xq27.3

Inheritance: X-linked triplet repeat

Clinical Features:

- Most common inherited causes of intellectual disability (ID)
- Delayed development, ID (mod-severe in boys, milder in girls), autistic features
- *Physical features:* long face, high forehead, large ears, prominent jaw, hyperextensible joints, mitral valve prolapse and macroorchidism (enlarged testes)
- *Carrier females:* anxiety, OCD, depression, 20% have **premature ovarian insufficiency** (compared to 1% in general population)
- *Carrier males:* progressive intention tremor, ataxia and parkinsonism (**Fragile X-associated tremor/ataxia syndrome (FXTAS)**)



Fragile X Syndrome (FXS)

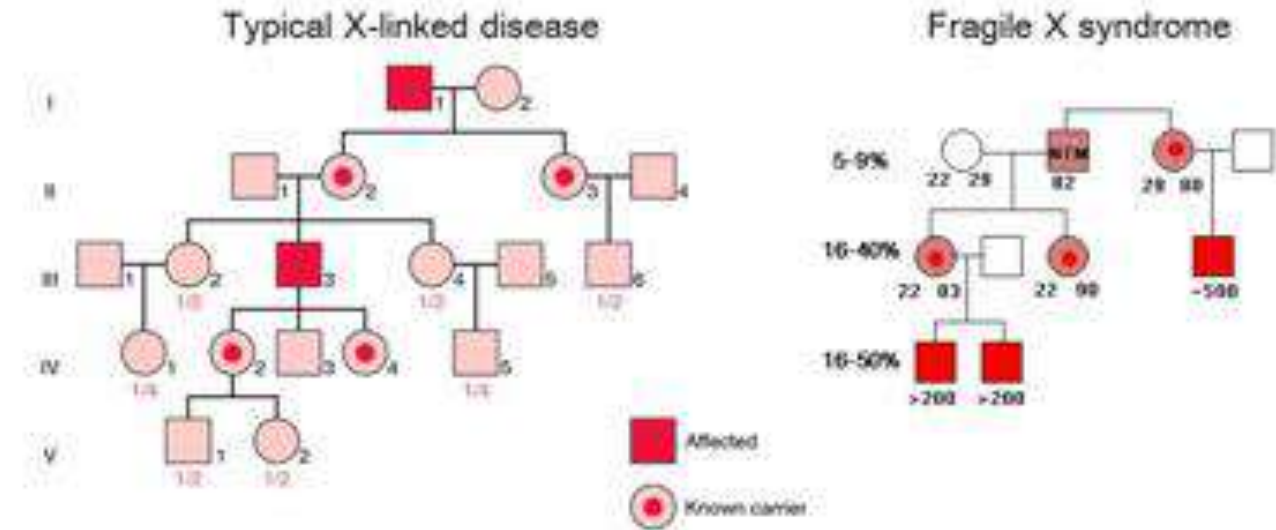
Molecular Tests:

Increased CGG triplet repeat

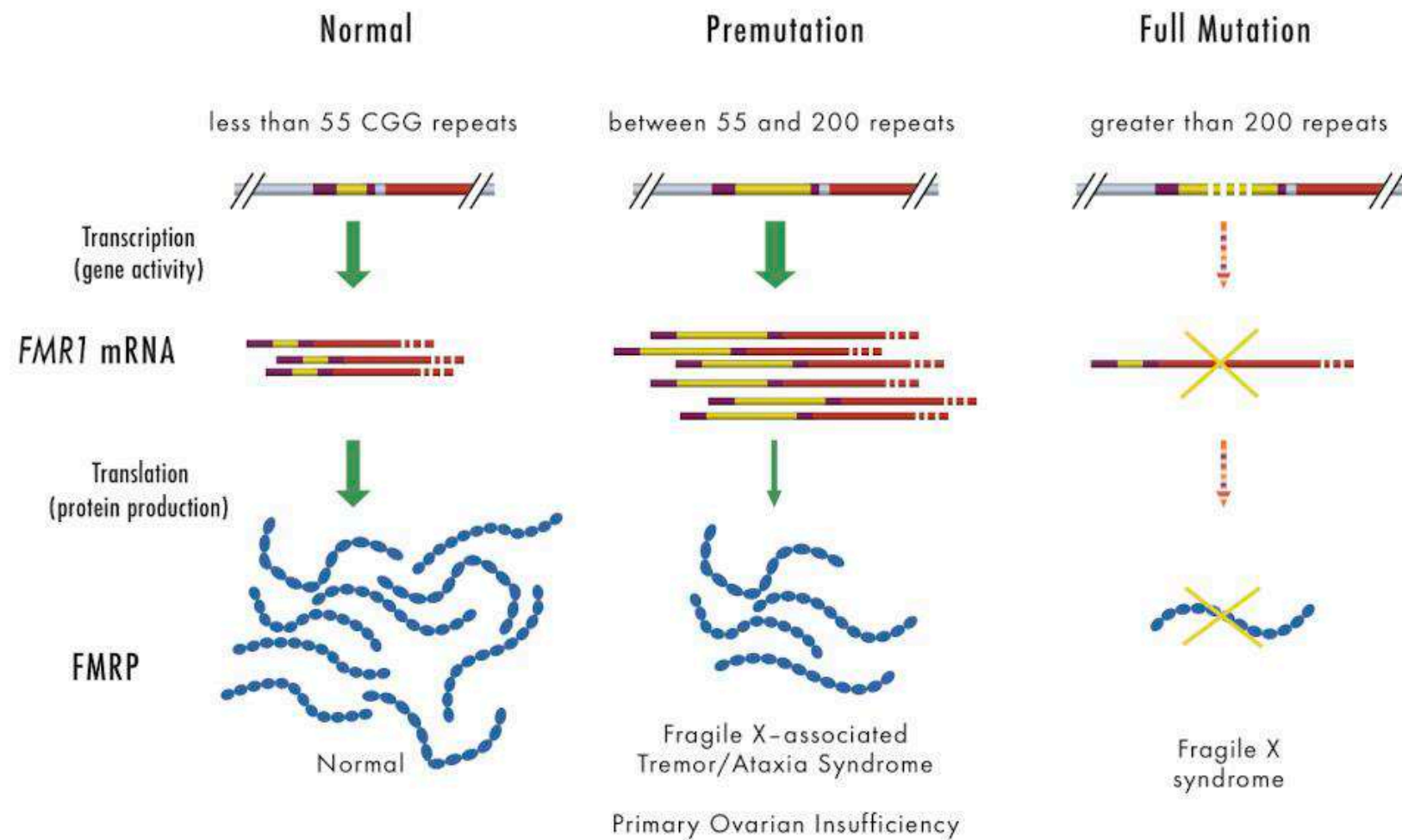
- Normal: 5-44 repeats
- Pre-mutation: 55-200 repeats
- Mutation: >200 repeats

Anticipation:

- FXS is a trinucleotide repeat disorder that may demonstrate anticipation
- Anticipation occurs when individuals with a premutation transmit unstable *FMR1* alleles to their offspring
- Example: transmission from a grandfather who carries a premutation to his daughter, whose premutation expands into a full mutation when she transmits it to her son



FMR1-Related Disorders





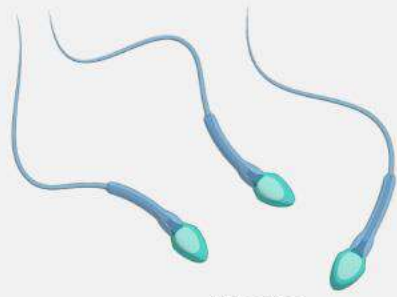
Mosaicism



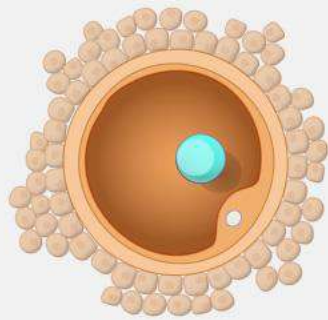
Germ line (germ cells)

Haploid

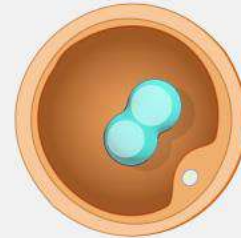
23 chromosomes (n) in human



Sperm



Ovum (egg)

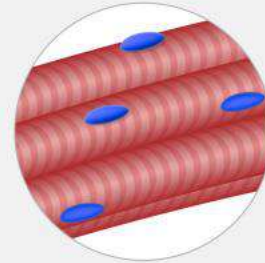


Fertilized egg

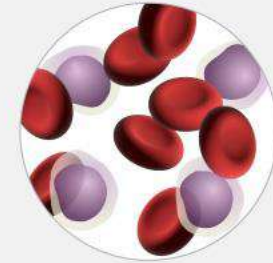
Somatic cells

Diploid

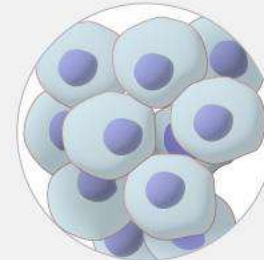
46 chromosomes (2n) in human



Skeletal and muscle cells



Blood cells



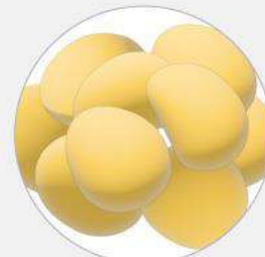
Stem cells



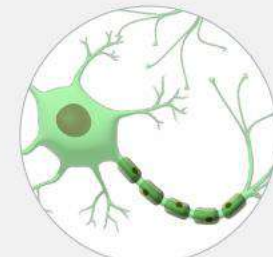
All other cells



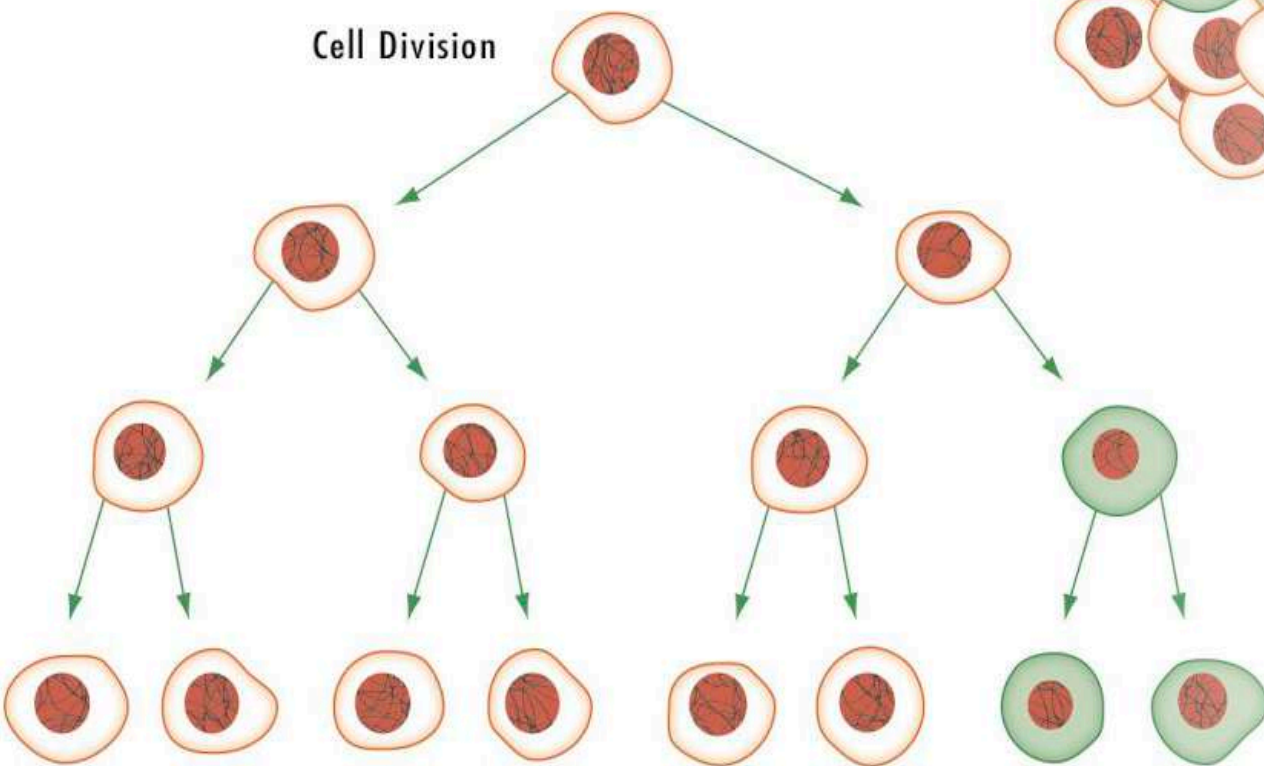
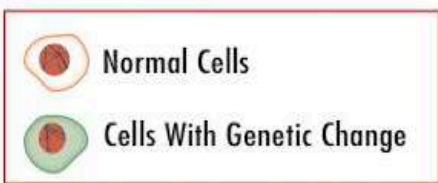
Organ and tissue cells



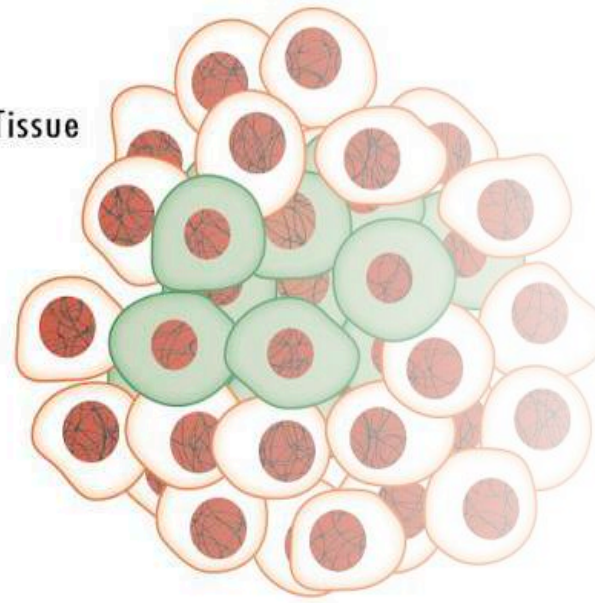
Fat cells



Neuron cells



Mosaic Tissue

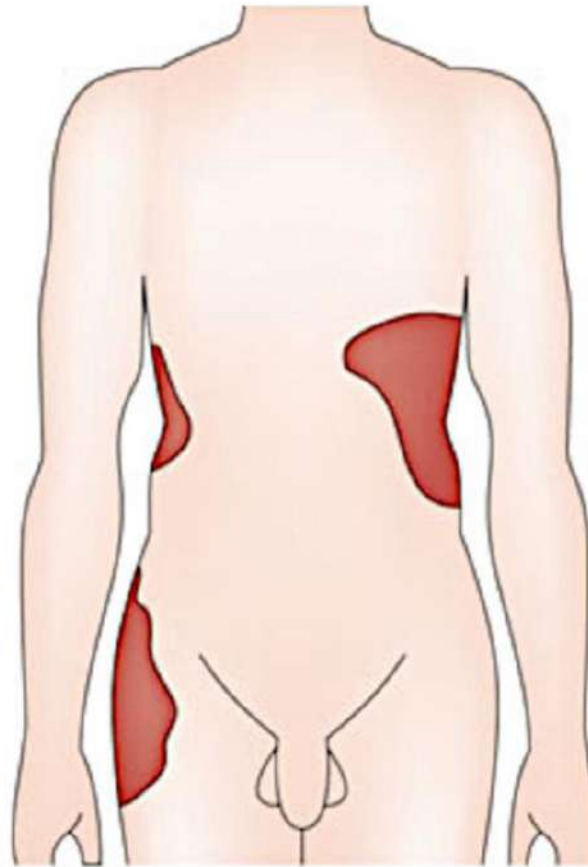


Mosaicism

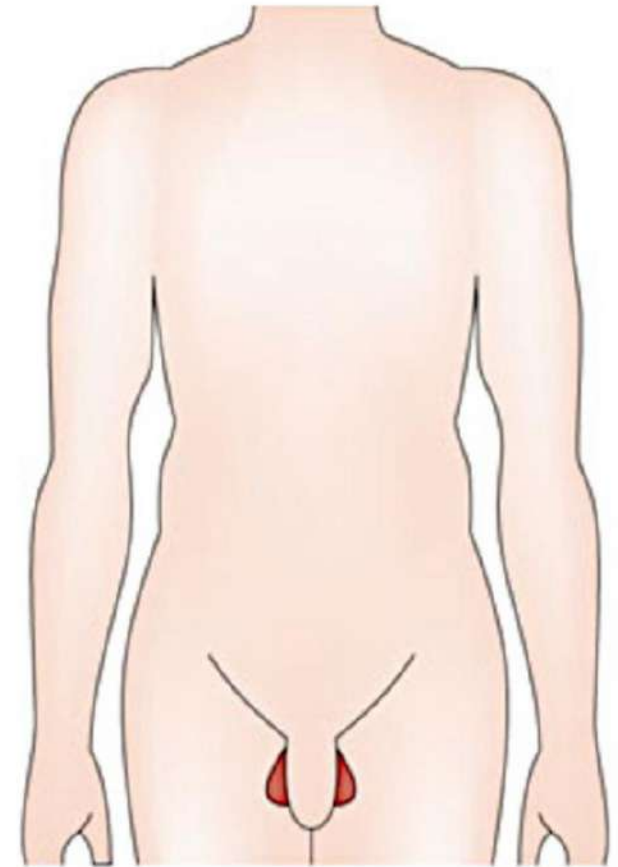
An individual, or a particular tissue of the body, can consist of **more than one cell type or line**, through an error occurring during **mitosis** at any stage after conception

Mosaicism

Mosaicism of either somatic tissues or germ cells can account for some instances of unusual patterns of inheritance or phenotypic features in an affected individual



Somatic Mosaicism

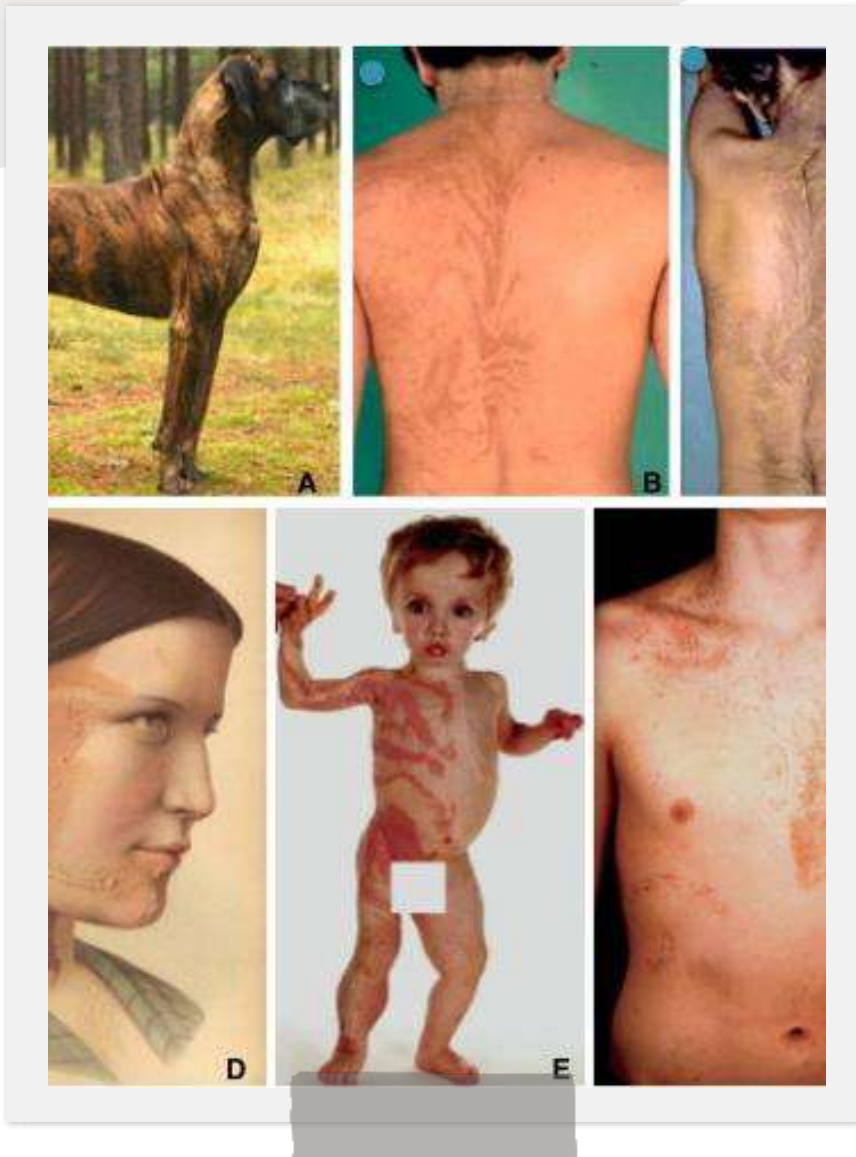


Germline Mosaicism

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Somatic Mosaicism

The possibility of somatic mosaicism is suggested by the features of a single-gene disorder being **less severe** in an individual than is usual, or by being **confined to a particular part of the body** in a segmental distribution

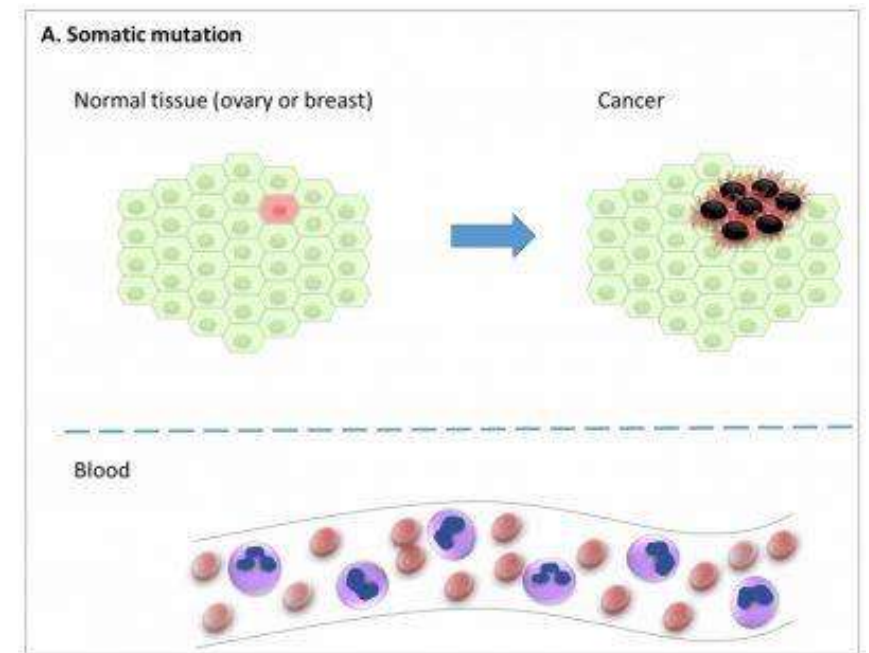


Somatic Mosaicism: Examples

- Mosaic neurofibromatosis type I
- Cancer is a classic example of somatic mosaicism
 - Since these mutations are only present in the cancer cells and not the rest of the cells in the body, the individual has somatic mosaicism for these mutations



García-Romero, M. T., Parkin, P., & Lara-Corrales, I. (2016). Mosaic neurofibromatosis type 1: a systematic review. *Pediatric dermatology*, 33(1), 9-17.



Lønning, P. E., Eikesdal, H. P., Løes, I. M., & Knappskog, S. (2019). Constitutional Mosaic Epimutations-a hidden cause of cancer?. *Cell Stress*, 3(4), 118.

Gonadal Mosaicism

When the parents are phenotypically normal, and the results of genetic tests are also normal, but they have more than one of their children affected with a genetic condition

→ the most favored explanation for these observations is gonadal, or germline, mosaicism in one of the parents, i.e., the variant is present in a proportion of the gonadal or germline cells

- e.g.; germline mosaicism occurs in AD & XLR disorders
- It is important to keep germline mosaicism in mind when providing recurrence risks in genetic counseling for apparently new autosomal dominant and X-linked recessive mutations



www.bohring-opitz.org

<https://bohring-opitz.org/2018/04/23/maternal-mosaicism/>

Summary

- Genomic Imprinting and Uniparental Disomy: PWS and AS
- Mitochondrial Inheritance
- Anticipation and Repeat Expansion
- Mosaicism: somatic vs germline mosaicism



Reference

Emery's Elements of Medical Genetics 16th
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